

Inner ear

The inner ear contains the organ of hearing and the organs of balance. All are located within the labyrinth, a series of interlinked cavities in the petrous temporal bone containing interconnected membranous sacs and ducts. All spaces within the labyrinth are filled with fluid. The different sacs contain sensory epithelia consisting of supporting cells and mechanosensory cells, the hair cells that underlie acoustico-lateralis sensory systems in all vertebrates. In humans, there are six such mechanosensory epithelia: the organ of Corti within the cochlea (the hearing organ); the utricle and saccule (static balance organs); and the cristae of the semicircular canals (dynamic balance organs). Whilst sharing the same basic structure, hair cells and the accessory systems that surround them show specific adaptations to each of the different sensory modalities.

The disarticulated temporal bone is described in detail in [Chapter 37](#). The internal acoustic meatus (internal auditory canal) and bony and membranous labyrinths are described here.

OSSEOUS (BONY) LABYRINTH

The bony labyrinth consists of the vestibule (sacculus and utriculus/sacculle and utricle), semicircular canals and cochlea, which are all cavities lined by periosteum and which contain the membranous labyrinth ([Fig. 38.1](#)). The bone is denser and harder than that of the other parts of the petrous bone, and it is therefore possible, particularly in young skulls, to dissect the bony labyrinth out from the petrous temporal bone.

The osseous and membranous labyrinths are filled with fluid ([Fig. 38.2](#)). The gap between the internal wall of the osseous labyrinth and the external surface of the membranous labyrinth is filled with perilymph, a clear fluid with an ionic composition similar to that of other extracellular fluids, i.e. low in potassium ions and high in sodium and calcium. The membranous labyrinth contains endolymph, a fluid with an ionic composition more like that of cytosol, i.e. high in potassium ions and low in sodium and calcium. Moreover, the endolymphatic compartment has an electrical potential that is approximately 80 mV more positive than the perilymphatic compartment (the endolymphatic potential). These differences in ionic composition and potential, maintained by homeostatic tissues in the walls of the labyrinth, are essential to maximize the sensitivity of the mechanosensory hair cells that convert the vibrations set up in the inner ear fluids by head or sound movements into electrical signals that are transmitted via the vestibulocochlear nerve to the vestibular and cochlear nuclei, respectively, in the brainstem.

VESTIBULE

The vestibule is the central part of the bony labyrinth and lies medial to the tympanic cavity, posterior to the cochlea and anterior to the semicircular canals (see [Fig. 38.1](#)). It is somewhat ovoid in shape but flattened transversely, and (on average) measures 5 mm from front to back and vertically, and 3 mm across. In its lateral wall is the opening of the oval window (fenestra vestibuli), into which the base of the stapes inserts, and to which the base of the stapes is attached by an annular ligament. Anteriorly, on the medial wall, is a small spherical recess that contains the saccule; it is perforated by several minute holes, the macula cribrosa media, which transmit fine branches of the vestibular nerve to the saccule. Behind the recess is an oblique vestibular crest, the anterior end of which forms the vestibular pyramid. This crest divides below to enclose a small depression, the cochlear recess, which is perforated by vestibulocochlear fascicles as they pass to the vestibular end of the cochlear duct. Posterosuperior to the vestibular crest, in the roof and medial wall of the vestibule, is the elliptical recess (see [Fig. 38.1B](#)), which contains the utricle. The pyramid and adjoining part of

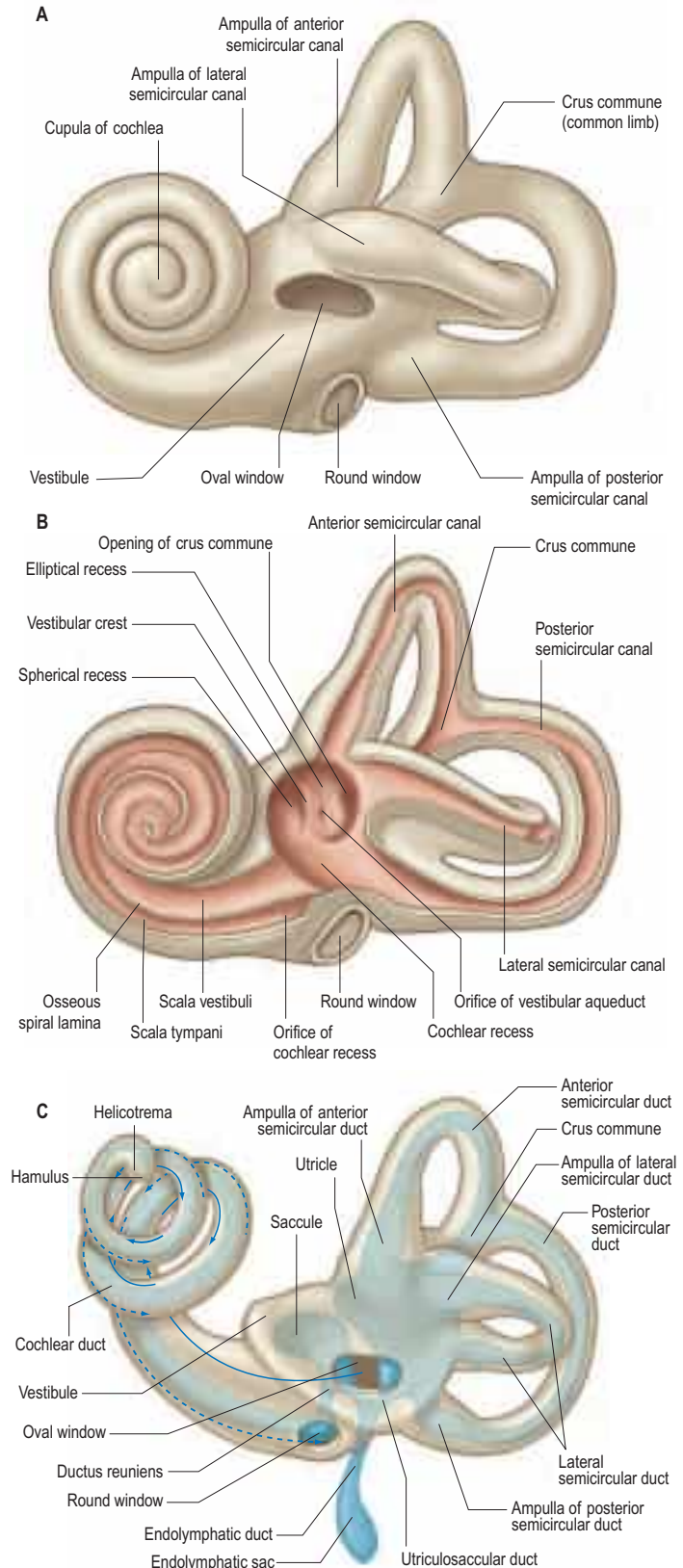


Fig. 38.1 The left bony labyrinth. **A**, Lateral aspect. **B**, Interior. **C**, The membranous labyrinth (blue) projected on to the bony labyrinth.

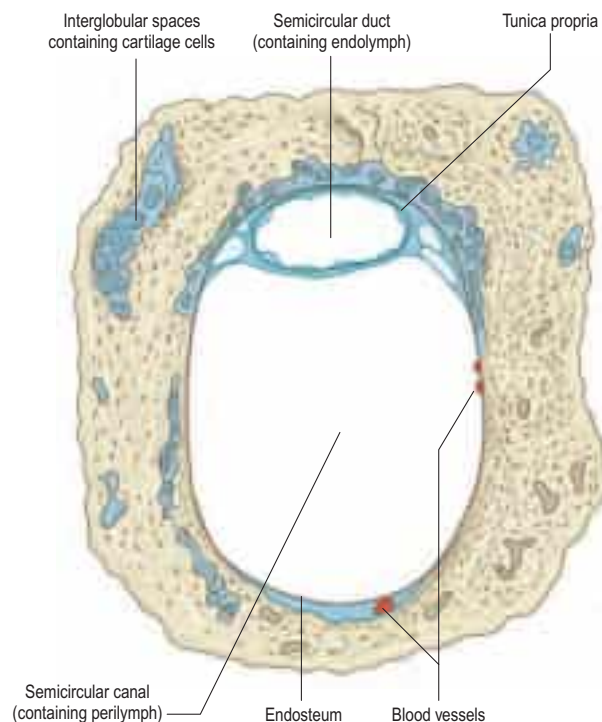


Fig. 38.2 A transverse section through the left posterior semicircular canal and duct.

the elliptical recess are perforated by a number of holes, the macula cribrosa superior; those in the former transmit nerves to the utricle and those in the latter transmit nerves to the ampullae of the anterior and lateral semicircular canals (see Fig. 38.1B). The region of the pyramid and elliptical recess corresponds to the superior vestibular area in the internal acoustic meatus (see Fig. 38.3). The vestibular aqueduct opens below the elliptical recess. It reaches the posterior surface of the petrous bone and contains one or more small veins and part of the membranous labyrinth, the endolymphatic duct (see Fig. 38.1C). In the posterior part of the vestibule are the five openings of the semicircular canals; in its anterior wall is an elliptical opening that leads into the scala vestibuli of the cochlea.

SEMICIRCULAR CANALS

The three semicircular canals – anterior (superior), posterior and lateral (horizontal) – are located posterosuperior to the vestibule (see Fig. 38.1). They are compressed from side to side and each forms approximately two-thirds of a circle. They are unequal in length but similar in diameter along their lengths, except where they bear a terminal swelling, an ampulla, which is almost twice the diameter of the canal.

The anterior semicircular canal is 15–20 mm long. It is vertical in orientation and lies transverse to the long axis of the petrous temporal bone under the anterior surface of its arcuate eminence. The eminence may not accurately coincide with this semicircular canal but may instead be adapted to the occipitotemporal sulcus on the inferior surface of the temporal lobe of the brain. The ampulla at the anterior end of the canal opens into the upper and lateral part of the vestibule. Its other end unites with the upper end of the posterior canal to form the crus commune (common limb), which is 4 mm long, and opens into the medial part of the vestibule.

The posterior semicircular canal is also vertical but curves backwards almost parallel with the posterior surface of the petrous bone. It is 18–22 mm long and its ampulla opens low in the vestibule, below the cochlear recess where the macula cribrosa inferior transmits nerves to it. Its upper end joins the crus commune.

The lateral canal is 12–15 mm long and its arch runs horizontally backwards and laterally. Its anterior ampulla opens into the upper and lateral angle of the vestibule, above the oval window and just below the ampulla of the anterior canal; its posterior end opens below the opening of the crus commune.

The two lateral semicircular canals of the two ears are often described as being in the same plane and the anterior canal of one side as being almost parallel with the opposite posterior canal. However, measurements of the angular relations of the planes of the semicircular osseous

canals in 10 human skulls led Blanks et al (1975) to suggest that the planes of the three ipsilateral canals are not completely perpendicular to each other. The angles were measured as: lateral/anterior $111.76 \pm 7.55^\circ$, anterior/posterior $86.16 \pm 4.72^\circ$, posterior/lateral $95.75 \pm 4.66^\circ$. The planes of similarly orientated canals of the two sides also showed some departure from being parallel: left anterior/right posterior $24.50 \pm 7.19^\circ$, left posterior/right anterior $23.73 \pm 6.71^\circ$, left lateral/right lateral $19.82 \pm 14.93^\circ$. The same observers (Curthoys et al 1977) also measured the dimensions of the canals. The mean radii of the osseous canals were found to be as follows: lateral 3.25 mm, anterior 3.74 mm, posterior 3.79 mm. The diameters of the osseous canals are 1 mm (minor axis) and 1.4 mm (major axis). The membranous ducts within them are much smaller, but are also elliptical in transverse section, and have major and minor axes of 0.23 and 0.46 mm (see Fig. 38.2). Representative means for ampullary dimensions are as follows: length 1.94 mm, height 1.55 mm. Phylogenetic studies suggest that the arc sizes of the semicircular canals in humans and other primates may be functionally linked to sensory control of body movements. The angulation and dimensions of the canals may be related to locomotor behaviour and possibly to agility, or more specifically to the frequency spectra of natural head movements (see review by Spoors and Zonneveld (1998)).

COCHLEA

The cochlea (from the Greek *cochlos* for snail) is the most anterior part of the labyrinth, lying in front of the vestibule (see Figs 38.1 and 38.9A). It is 5 mm from base to apex, and 9 mm across its base. Its apex, or cupula, points towards the anterosuperior area of the medial wall of the tympanic cavity (see Fig. 38.9A). Its base faces the bottom of the internal acoustic meatus and is perforated by numerous apertures for the cochlear nerve. The cochlea has a conical central bony core, the modiolus, and a spiral canal runs around it. A delicate osseous spiral lamina (or ledge) projects from the modiolus, partially dividing the canal (see Fig. 38.9B). Within this bony spiral lies the membranous cochlear duct, attached to the modiolus at one edge and to the outer cochlear wall by its other edge. There are therefore three longitudinal channels within the cochlea. The middle canal (the cochlear duct or scala media) is blind and ends at the apex of the cochlea; its flanking channels communicate with each other at the modiolar apex at a narrow slit, the helicotrema (see Fig. 38.1C). Two elastic membranes form the upper and lower bounds of the scala media. One is Reissner's membrane, the thin vestibular membrane that separates the scala media from the scala vestibuli. The other is the basilar membrane, which forms the partition between the scala media and the scala tympani. The organ of Corti, the sensory epithelium of hearing, sits on the inner surface of the basilar membrane. At the base of the scala vestibuli is the oval window (fenestra vestibuli), which leads on to the vestibular cavity but is sealed by the footplate of the stapes. The scala tympani is separated from the tympanic cavity by the secondary tympanic membrane at the round window (fenestra cochleae). The central cochlear core, the modiolus, has a broad base near the lateral end of the internal acoustic meatus, where it corresponds to the spiral tract (tractus spiralis foraminosus). There are several openings in this area for the fascicles of the cochlear nerve: those for the first $1\frac{1}{2}$ turns run through the small holes of the spiral tract, and those for the apical turn run through the hole that forms the centre of the tract. Canals from the spiral tract go through the modiolus and open in a spiral sequence into the base of the osseous spiral lamina. Here the small canals enlarge and fuse to form Rosenthal's canal, a spiral canal in the modiolus that follows the course of the osseous spiral lamina and contains the spiral ganglion (see Fig. 38.9B). The main tract continues through the centre of the modiolus to the cochlear apex.

The osseous cochlear canal spirals for about $2\frac{3}{4}$ turns around the modiolus and is 35 mm long. At its first turn, the canal bulges towards the tympanic cavity, where it underlies the promontory. At the base of the cochlea, the canal is 3 mm in diameter but it becomes progressively reduced in diameter as it spirals apically to end at the cupula. In addition to the round and oval windows, which are the two main openings at its base, the canal has a third, smaller opening for the cochlear aqueduct or canaliculus. The latter is a minute funnel-shaped canal that runs to the inferior surface of the petrous temporal bone; it transmits a small vein to the inferior petrosal sinus (see p. 437) and connects the sub-arachnoid space to the scala tympani.

The osseous or primary spiral lamina is a ledge that projects from the modiolus into the osseous canal like the thread of a screw (see Fig. 38.9B). It is attached to the inner edge of the basilar membrane and ends in a hook-shaped hamulus at the cochlear apex, partly bounding

the helicotrema (see Fig. 38.1C). From Rosenthal's canal, many tiny canals, the habenula perforata, radiate through the osseous lamina to its rim, where they each carry a fascicle of the cochlear nerve through the foramen nervosum to the organ of Corti. A secondary spiral lamina projects inwards from the outer cochlear wall towards the osseous spiral lamina and is attached to the outer edge of the basilar membrane. It is most prominent in the lower part of the first turn; the gap between the two laminae increases progressively towards the cochlear apex, which means that the basilar membrane is wider at the apex of the cochlea than at the base.

MICROSTRUCTURE OF THE BONY LABYRINTH

The wall of the bony labyrinth is lined by fibroblast-like perilymphatic cells and extracellular matrix fibres (see Fig. 38.2). The morphology of the cells varies in different parts of the labyrinth. Where the perilymphatic space is narrow, as in the cochlear aqueduct, the cells are reticular or stellate in form; they give off sheet-like cytoplasmic extensions that cross the extracellular space. Where the space is wider, as in the scala vestibuli and tympani of the cochlea and much of the vestibule, the perilymphatic cells on the periosteum and the external surface of the membranous labyrinth are extremely flat and resemble a squamous epithelium. Elsewhere, on parts of the perilymphatic surface of the basilar membrane, the cells are cuboidal.

Recent evidence suggests that micropores or canaliculi (canaliculi perforantes) (0.2–23.0 μm diameter) are more widely distributed within the bony surfaces lining the perilymphatic space than was previously suspected; they are numerous in the peripheral and modiolar portions of the osseous spiral lamina and the floor of the scala tympani, but sparse in the osseous wall of the scala vestibuli. The proposal that these canaliculi normally provide an extensive fluid communication channel between the scala tympani and the spiral canal of the cochlea could have implications not only for novel drug-based cochlear therapies delivered via the scala tympani and the delivery of stem cells or appropriate cell lines into the deafened cochlea, but also for the design of implanted perimodiolar electrode arrays (Shepherd and Colreavy 2004). (For further reading about the changes in the inner ear that are induced by implanted cochlear electrodes, both acute and long-term, see Keifer et al (2006).)

Composition of inner ear fluids

Perilymph resembles cerebrospinal fluid in ionic composition, particularly in the scala tympani. Its composition differs a little between the two cochlear scalae: concentrations of potassium, glucose, amino acids and proteins are greater in the scala vestibuli. This has led to the suggestion that perilymph in the scala vestibuli is derived from plasma via the endothelial boundary of the cochlear blood vessels, whereas the perilymph in the scala tympani contains some cerebrospinal fluid derived from the subarachnoid spaces via the cochlear canaliculus. However, the lack of significant bulk flow suggests that perilymph homeostasis is predominantly locally regulated. Perilymph contains approximately 5 mM K^+ , 150 mM Na^+ , 120 mM Cl^- and 1.5 mM Ca^{2+} .

The membranous labyrinth is filled with endolymph, a fluid produced by the marginal cells of the stria vascularis and the dark cells of the vestibule (see review by Wangemann and Schacht (1996)) (see Fig. 38.2B). Whatever their relative contributions, endolymph probably circulates in the labyrinth; it enters the endolymphatic sac, where it is transferred into the adjacent vascular plexus via the specialized epithelium of the sac. Pinocytotic removal of fluid may also occur in other labyrinthine regions.

Endolymph contains greater K^+ (150 mM) and Cl^- (130 mM) concentrations and lower Na^+ (2 mM) and Ca^{2+} (20 μM) concentrations than perilymph. The high potassium concentration is important for the function of the mechanosensory hair cells and is maintained by the actions of the lateral wall, which contains two tissues, namely: the spiral ligament and the stria vascularis. Together, these tissues promote the recirculation of potassium from perilymph back to endolymph by uptake via potassium channels and gap junctional communication. Gap junctions are formed from connexins; their importance to this process is emphasized by the fact that mutations in connexins are significant causes of hearing loss (Duman and Tekin 2013). The vestibular regions may not have an endolymphatic potential, as their lateral wall structure is simplified compared to that of the cochlea, although the difference in potassium concentration between endolymph and perilymph remains important.

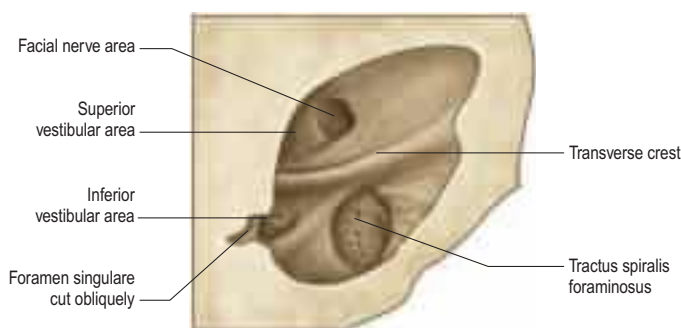


Fig. 38.3 The fundus of the left internal acoustic meatus, exposed by a section through the petrous part of the left temporal bone nearly parallel to the line of its superior border.

INTERNAL ACOUSTIC MEATUS

The internal acoustic meatus (internal acoustic/auditory canal) is separated from the internal ear at its lateral fundus by a vertical plate divided unequally by a transverse (falciform) crest (Fig. 38.3). Five nerves – facial, nervus intermedius, cochlear, superior and inferior vestibular – pass through openings in the vertical plate, above and below the transverse crest. The facial and superior vestibular nerves enter canals that are superior to the crest. The facial nerve is anterior to the superior vestibular nerve, from which it is separated at the lateral end of the meatus by a vertical ridge of bone (Bill's bar). The nervus intermedius lies between the facial motor root and the superior vestibular nerve, to which it may be adherent. The superior vestibular area contains openings for nerves to the utricle and anterior and lateral semicircular ducts. Below the crest, an anterior cochlear area contains a spiral of small holes, the tractus spiralis foraminosus, which encircles the central cochlear canal. Behind this, the inferior vestibular area contains openings for saccular nerves, and most posteroinferiorly, a single hole (foramen singulare) admits the nerve to the posterior semicircular duct. It has been suggested that vascular loops in the internal acoustic meatus (from the anterior inferior cerebellar artery) might generate pulsatile tinnitus.

MEMBRANOUS LABYRINTH

The membranous labyrinth is separated from the periosteum by a space that contains perilymph and a web-like network of fine blood vessels (see Figs 38.1C, 38.2). It can be divided into two major regions: the vestibular apparatus and the cochlear duct.

VESTIBULAR APPARATUS

The vestibular apparatus consists of three membranous semicircular canals that communicate with the utricle, a membranous sac leading into a smaller chamber, the saccule, via the utriculosaccular duct. This Y-shaped duct has a side branch to the endolymphatic duct, which passes to the endolymphatic sac, a small but functionally important expansion situated under the dura of the petrous temporal bone. From the saccule, a narrow canal, the ductus reuniens, leads to the base of the cochlear duct. These various ducts and sacs form a closed system of intercommunicating channels. Endolymph is resorbed into the cerebrospinal fluid from the endolymphatic sac, which therefore provides the site for the drainage of endolymph for the entire membranous labyrinth.

The terminal fibres of the vestibular nerve are connected to the five specialized sensory epithelia (two maculae and three cristae) in the walls of the membranous labyrinth. Maculae are flat plaques of sensory hair cells surrounded by supporting cells, and are found in the utricle and saccule. The cristae (crests) are ridges bearing sensory hair cells and supporting cells. They are found in the walls of the ampullae near the utricular openings of the three semicircular canals, one for each canal.

Utricle

The utricle is the larger of the two major vestibular sacs. It is an irregular, oblong, dilated sac that occupies the posterosuperior region of the vestibule (see Fig. 38.1C), and contacts the elliptical recess (where it is a blind-ended pouch) and the area inferior to it.

The macula of the utricle (or utriculus) is a specialized neurosensory epithelium lining the membranous wall, and is the largest of the vestibular sensory areas (Fig. 38.4). It is triangular or heart-shaped in surface view and lies horizontally with its long axis orientated antero-posteriorly and its sharp angle pointing posteriorly (Fig. 38.5). It is flat except at the anterior edge, where it is gently folded in on itself, and it measures 2.8 mm long by 2.2 mm wide. The mature form of the macula is reached early in development, but in the adult a bulge is often present on the anterolateral border; there is sometimes an indentation at the anteromedial border. The epithelial surface is covered by the otolithic membrane (statoconial membrane), a gelatinous structure in which many small crystals, the otoconia (otoliths, statoliths), are embedded. A curved ridge, the 'snowdrift line', runs along the length of the otolithic membrane. It corresponds to a narrow crescent of underlying sensory epithelium termed the striola, 0.13 mm wide. The density of sensory hair cells in this strip of epithelium is 20% less than in the rest of the macula. The striola is convex laterally and runs from the medial aspect of the anterior margin in a posterior direction towards, but not reaching, the posterior pole. The part of the macula medial to the striola is called the pars interna and is slightly larger than the pars externa, which is lateral to it. The significance of this area is that the sensory cells are functionally and anatomically polarized with respect to the

midline of the striola (see Fig. 38.5). The macula in each utricle is approximately horizontal when the head is in its normal position. Linear acceleration of the head in any horizontal plane will result in the otolithic membrane lagging behind the movement of the membranous labyrinth as a result of the inertia produced by its mass. The membrane thus maximally stimulates one group of hair cells by deflecting their bundles towards the striola whilst inhibiting others by deflecting their bundles away from it. Hence each horizontal movement of the head will produce a specific pattern of firing in utricular afferent nerve fibres.

Saccul

The saccul (or sacculus) is a slightly elongated, globular sac lying in the spherical recess near the opening of the scala vestibuli of the cochlea (see Fig. 38.1C). The saccular macula is an almost elliptical structure, 2.6 mm long and 1.2 mm at its widest point. Its long axis is orientated anteroposteriorly but, in contrast to the utricular macula, the saccular macula lies in a vertical plane on the wall of the saccul. Its elliptical shape is very slightly distorted by a small anterosuperior bulge. Like the utricular macula, it is covered by an otolithic (statoconial) membrane and possesses a striola similar to that of the utricle, 0.13 mm wide, which extends along its long axis as an S-shaped strip about which the sensory cells are functionally and anatomically polarized (see Fig. 38.5). The part of the macula above the striola is termed the pars interna, and that below it, the pars externa. The operation of the saccul is similar to that of the utricle. However, because of its vertical orientation, the saccul is particularly sensitive to linear acceleration of the head in the vertical plane and is, therefore, a major gravitational sensor when the head is in an upright position. It is also particularly sensitive to movement along the anteroposterior axis.

Semicircular canals

The lateral, anterior and posterior semicircular ducts follow the course of their osseous canals. Throughout most of their length they are securely attached, by much of their circumference, to the osseous walls. They are approximately one-quarter of the diameter of their osseous canals (see Fig. 38.2). The medial ends of the anterior and posterior canals fuse to form a single common duct, the crus commune, before entering the utricle. The lateral end of each canal is dilated to form an ampulla, within the ampulla of the osseous canal. The short segment of duct between the ampullae and utricle is the crus ampullaris.

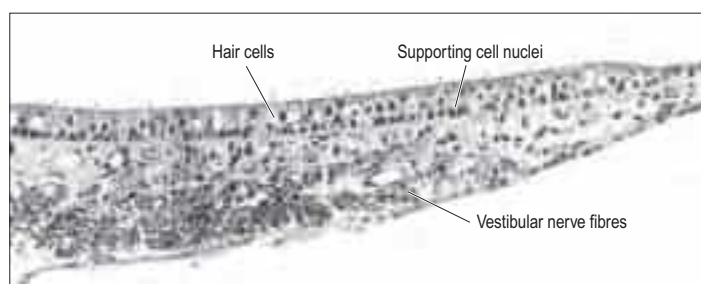


Fig. 38.4 A section of the utricular macula from a guinea pig, showing the relative positions of the hair cells and supporting cell nuclei. Semi-thin resin section, toluidine blue stain. (The inner ear is extremely vulnerable to hypoxia and situated in one of the hardest bones in the body, which means that well-fixed human tissue is rarely obtained for histology. Guinea pigs are one of the most frequently used animal models of human hearing and their inner ear ultrastructure is very similar.) (Courtesy of RM Walsh, DN Furness and CM Hackney, The Institute of Science and Technology in Medicine, School of Life Sciences, Keele University.)

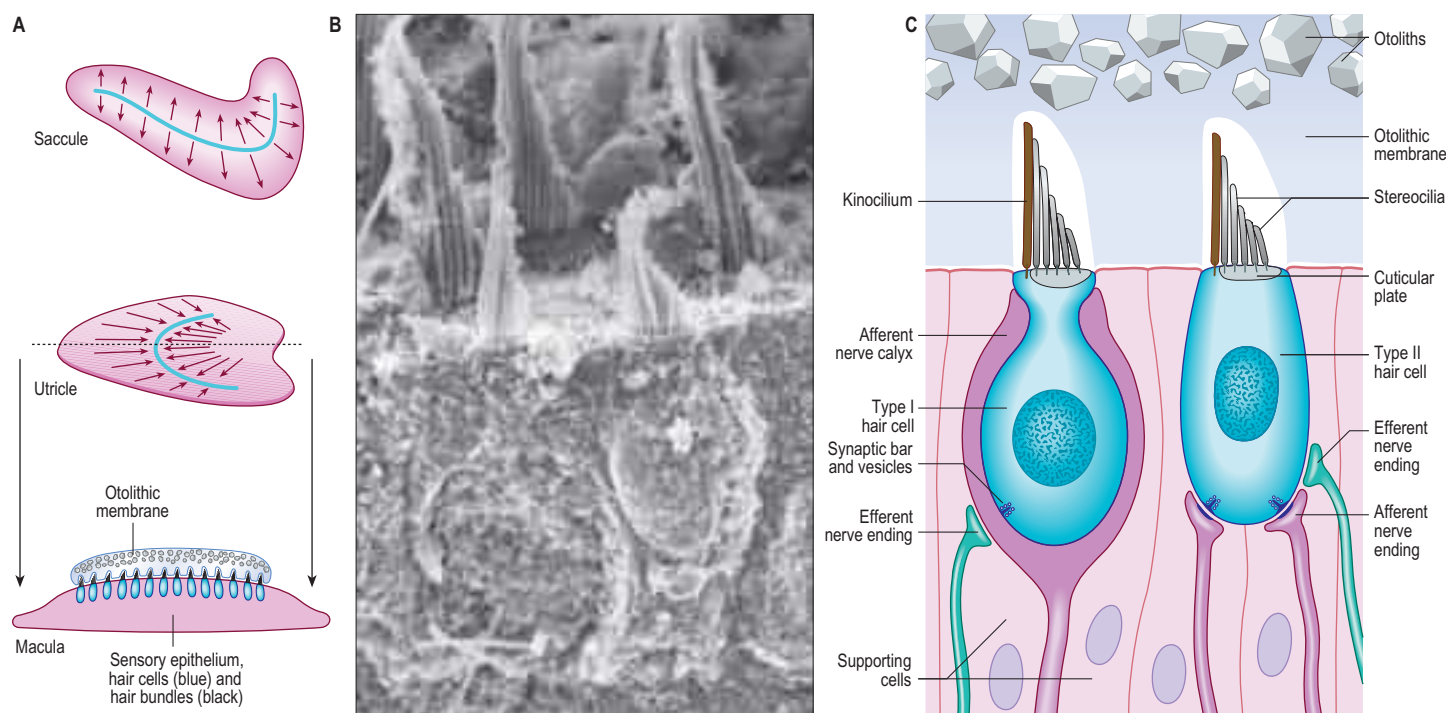


Fig. 38.5 **A**, The morphological organization of the saccular and utricular maculae and the relationship of their hair cells to the otolithic membrane. The utricular macula has been tilted in the plane of the page to emphasize that it lies horizontally, whereas the saccular macula lies vertically when the head is in an upright position. Note the different shapes of the maculae, the position of the striola as indicated by a curved line in each case, and the different orientational orientations of their stereociliary bundles. The arrows indicate the excitatory direction of deflection. **B**, A scanning electron micrograph of a fracture of a utricular macula (guinea pig) showing a type I hair cell (left) and a type II hair cell (right). **C**, The differing innervation patterns of the two types of hair cell. (B, Courtesy of DN Furness, The Institute of Science and Technology in Medicine, School of Life Sciences, Keele University.)

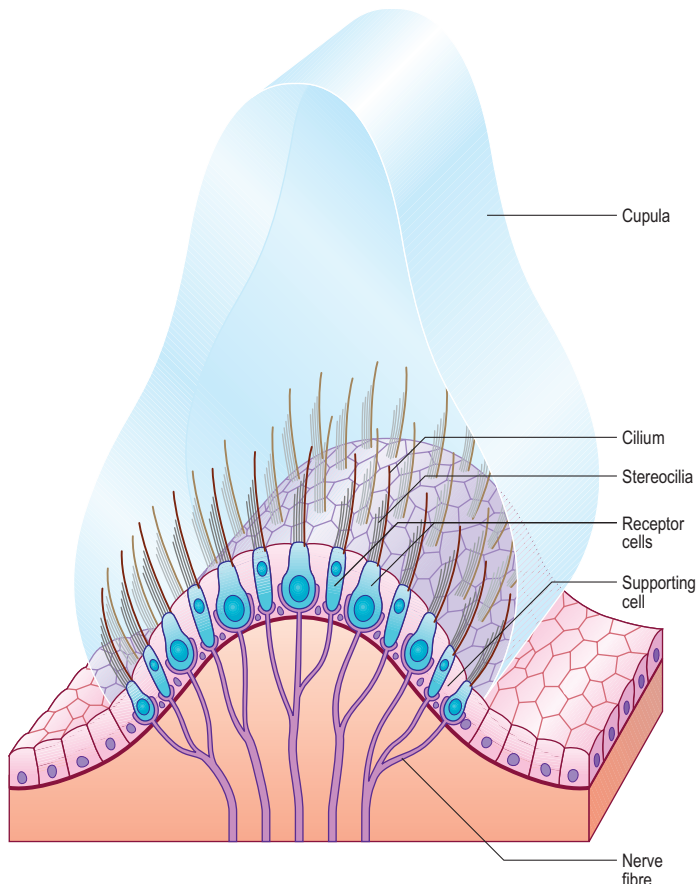


Fig. 38.6 A section of an ampullary crest.

The membranous wall of each ampulla contains a transverse elevation (septum transversum), on the central region of which is a saddle-shaped sensory ridge, the ampullary crest, containing hair cells and supporting cells. It is broadly concave on its free edge along most of its length and has a concave gutter (planum semilunatum) at either end between the ridge and the duct wall. Sectioned across the ridge, the crests of the lateral and anterior semicircular canals have smoothly rounded corners; the posterior crest is more angular. A vertical plate of gelatinous extracellular material, the cupula, is attached along the free edge of the crest (**Fig. 38.6**). It projects far into the lumen of the ampulla so that it is readily deflected by movements of endolymph derived from head rotations within the duct, by means of which stimuli are delivered to the sensory hair cells. The three semicircular canals thus detect angular accelerations during tilting or turning movements of the head in all three different planes of three-dimensional space.

Microstructure of the vestibular system

The maculae and crests detect the orientation of the head with respect to gravity and changes in head movement by means of the mechano-sensitive hair cells. These hair cells are in synaptic contact with afferent and efferent endings of the vestibular nerve on their basolateral aspect. The entire epithelium lies on a bed of thick, fibrous connective tissue containing myelinated vestibular nerve fibres and blood vessels. The axons lose their myelin sheaths as they perforate the basal lamina of the sensory epithelium. There are two types of sensory hair cell in the vestibular system, type I and type II (see **Figs 38.5, 38.8**).

Type I vestibular sensory cells measure 25 μm in length, with a free surface of 6–7 μm in diameter. The basal part of the cell does not reach the basal lamina of the epithelium. Each cell is typically bottle-shaped, with a narrow neck and a rather broad, rounded basal portion containing the nucleus (see **Fig. 38.5**). The apical surface is characterized by 30–50 stereocilia (large, regularly arranged, modified microvilli about 0.25 μm across) and a single kinocilium (with the typical '9 + 2' arrangement of microtubules characteristic of true cilia). The kinocilium is considerably longer than the stereocilia, and may attain 40 μm , whereas the stereocilia are of graded lengths. They are characteristically arranged in regular rows behind the kinocilium in descending order of height, the longest being next to the kinocilium (**Fig. 38.7**). The kinocilium emerges basally from a typical basal body, with a centriole immediately beneath it.

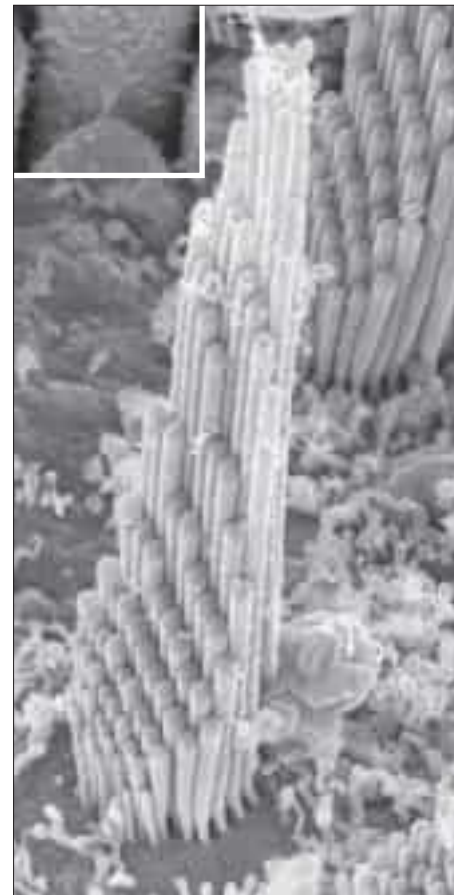


Fig. 38.7 A scanning electron micrograph of a stereociliary bundle from the utricle (guinea pig). The stereocilia are arranged in rows of increasing height towards the tallest element, the kinocilium. Deflection in the direction of the kinocilium results in depolarization of the hair cell. The inset shows a tip link connecting a short stereocilium tip to the tall stereocilium side behind. (Courtesy of DN Furness, The Institute of Science and Technology in Medicine, School of Life Sciences, Keele University.)

Close to the inner surface of their basal two-thirds, every cell contains numerous synaptic ribbons with associated synaptic vesicles. The postsynaptic surface of an afferent nerve ending encloses the greater part of the sensory cell body in the form of a cup (chalice or calyx). Efferent nerve fibres make synapses with the external surface of the calyx, rather than directly with the sensory cell.

The kinocilium confers structural polarity on the bundle, which relates to functional polarity. The stereocilia and kinocilium are all interconnected by fine extracellular filaments of various types, called cross links. One in particular, the tip link, connects the shorter stereocilia in each row with adjacent stereocilia in the taller row next to it (see **Fig. 38.7**). The tip link is common to all types of hair cell and is thought to play a central role in transduction; mutations in the proteins that comprise the tip link are significant in Usher syndrome, which is characterized by auditory and visual abnormalities. Deflection of the bundle towards the kinocilium results in depolarization of the hair cell and increases the rate of neurotransmitter release from its base. Deflection away from the kinocilium hyperpolarizes the hair cell and reduces the release of neurotransmitter. How deflections produce these responses will be considered in more detail later.

There is much greater variation in the sizes of type II sensory cells (see **Fig. 38.5B,C; Fig. 38.8**). Some are up to 45 μm long and almost span the entire thickness of the sensory epithelium, whereas others are shorter than type I cells. They are mostly cylindrical, but otherwise resemble type I cells in their contents and the presence of an apical kinocilium and stereocilia. However, their kinocilia and stereocilia tend to be shorter and less variable in length. The most striking difference between type I and II cells is their efferent nerve terminals: type II cells receive several efferent nerve boutons containing a mixture of small clear and dense-core vesicles around their bases, and afferent endings are small expansions rather than chalices.

Polarization allows the hair cells to have specific orientations that optimize their function within each sensory organ. In the maculae, they are arranged symmetrically on either side of the striola. In the utricle, the kinocilia are positioned on the side of the sensory cell nearest to the striola so that the excitatory direction is towards the midline. In the saccule, the structural and functional polarity is the opposite, i.e. away from it. In the ampullary crests, the cells are orientated with their rows of stereocilia at right angles to the long axis of the semicircular duct. In the lateral crest, the kinocilia are on the side towards the utricle, whereas in the anterior and posterior crests they are away from it. These different arrangements are important functionally because any given acceleration

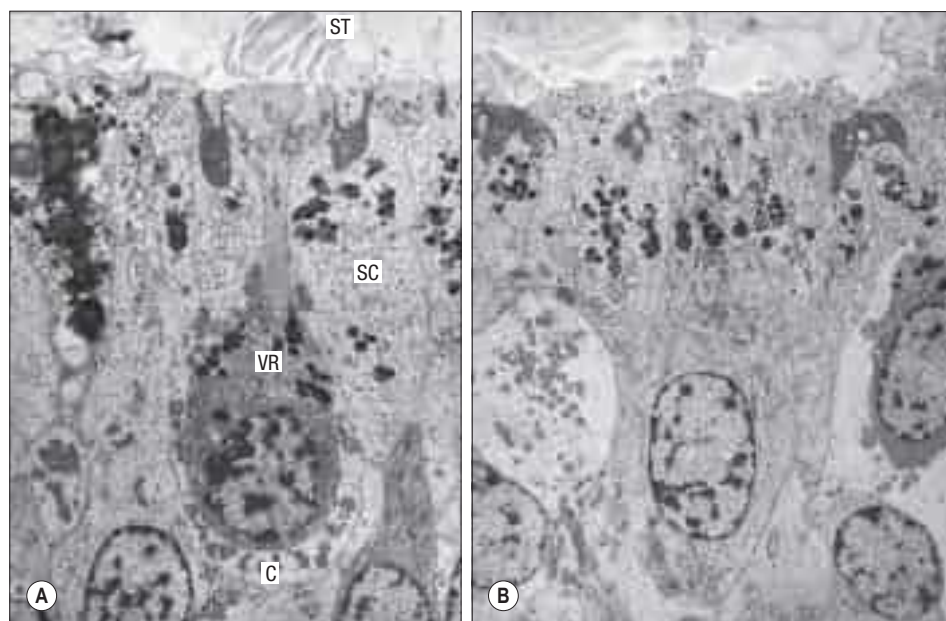


Fig. 38.8 Human vestibular hair cells (transmission electron micrographs). **A**, Type I cell (VR) bearing an apical group of stereocilia (ST) seen in a vertical section through the macula. Note that the hair cell is bottle-shaped, and that much of it is enclosed in the calyceal ending (C) of an afferent nerve terminal. Other abbreviations: SC, supporting cells. **B**, Human type II vestibular hair cell. A bouton-type afferent nerve terminal is in contact with the basal part. (Courtesy of H Felix, M Gleeson and L-G Johnsson, ENT Department, University of Zurich and GKT School of Medicine, London.)

of the head maximally depolarizes one group of hair cells and maximally inhibits a complementary set, thus providing a unique representation of the magnitude and orientation of any movement (for further details, see [Furness \(2002\)](#)).

The type I and II sensory cells are set within a matrix of supporting cells that reach from the base of the epithelium to its surface and form rosettes round the sensory cells, as seen in surface view. Although their form is irregular, they can easily be recognized by the position of their nuclei, which tend to lie below the level of sensory cell nuclei and just above the basal lamina (see [Fig. 38.4](#)). The apices of the supporting cells are attached by tight junctions to neighbouring supporting cells and to the hair cells to produce the reticular lamina, a composite layer that forms a plate that is relatively impermeable to cations other than via the mechanosensitive transduction channels of the hair cells.

The otolithic membrane is a layer of extracellular material divided into two strata. The external layer is composed of otoliths or otoconia, which are barrel-shaped crystals of calcium carbonate with angular ends, up to 30 μm long, and heterogeneous in distribution. They are attached to a more basal gelatinous layer into which the stereocilia and kinocilia of the sensory cells are inserted (see [Fig. 38.5](#)). The gelatinous material consists largely of glycosaminoglycans associated with fibrous protein.

Epley's manoeuvre

Benign paroxysmal positional vertigo is a condition in which a sensation of rotation with associated nystagmus is induced by adopting a particular position (with the abnormal ear dependent). It is believed that calcium carbonate crystals from the otoliths become freed from the otolithic membrane and, in certain positions, drop into the ampulla of the posterior semicircular canal, possibly becoming adherent to the cupula and rendering it gravity-sensitive. In certain positions, the alignment of the axis of the posterior semicircular canal with gravity results in the displacement of the cupula and the activation of the vestibulo-ocular reflex, leading to compensatory nystagmoid eye movements in response to apparent head movements.

Epley's canalith repositioning procedure relies on the adoption of a series of body postures designed to allow the aberrant crystals (or canaliths) to float out of the posterior semicircular canal and to stick to the wall of the vestibule. Cure rates in excess of 80% have been recorded and the procedures have largely superseded surgical procedures designed to denervate the ampulla of the posterior semicircular canal (singular neurectomy) or obliterate the canal completely.

Endolymphatic duct and sac

The endolymphatic duct runs in the osseous vestibular aqueduct and becomes dilated distally to form the endolymphatic sac. This is a structure of variable size, which may extend through an aperture on the posterior surface of the petrous bone to end between the two layers of the dura on the posterior surface of the petrous temporal bone near the sigmoid sinus (see [Fig. 38.1C](#)). The surface cells throughout the entire endolymphatic duct resemble those lining the non-specialized parts of the membranous labyrinth and consist of squamous or low cuboidal

epithelium. The epithelial lining and subepithelial connective tissue become more complex where the duct dilates to form the endolymphatic sac. An intermediate or rugose segment and a distal sac can be distinguished. In the intermediate segment, the epithelium consists of light and dark cylindrical cells. Light cells are regular in form and have numerous long surface microvilli with endocytic invaginations between them and large clear vesicles in their apical region. In contrast, dark cells are wedge-shaped and have a narrow base, few apical microvilli and dense, fibrillar cytoplasm.

The endolymphatic sac has important roles in the maintenance of vestibular function. Endolymph produced elsewhere in the labyrinth is absorbed in this region, probably mainly by the light cells. Damage to the sac, or blockage of its connection to the rest of the labyrinth, causes endolymph to accumulate; this produces hydrops, which affects both vestibular and cochlear function. The epithelium is also permeable to leukocytes, including macrophages, which can remove cellular debris from the endolymph, and to various cells of the immune system that contribute antibodies to this fluid.

COCHLEAR DUCT

The cochlear duct is a spiral tube that runs within the bony cochlea (see [Figs 38.1C, Fig. 38.9](#)). The osseous spiral lamina projects for part of the distance between the modiolus and the outer wall of the cochlea and is attached to the inner edge of the basilar membrane. Above it is a thickened ridge of endosteum called the spiral limbus. The endosteum of the outer wall is thickened to form a spiral cochlear ligament that projects inwards as a triangular basilar crest attached to the outer rim of the basilar membrane. Immediately above this is a concavity, the external spiral sulcus (sulcus spiralis externus), above which the thick, highly vascular periosteum projects as a spiral prominence. Above the prominence is a specialized, thick epithelial layer, the stria vascularis. A second, thinner vestibular membrane, Reissner's membrane, extends from the spiral limbus to the outer wall of the cochlea, where it is attached above the stria. Reissner's membrane consists of two layers of squamous epithelial cells separated by a basal lamina. The side facing the scala vestibuli bears flattened perilymphatic cells, with tight junctions between them, creating a diffusion barrier. The endolymphatic side is lined by squamous epithelial cells with many microvilli; these are also joined by tight junctions and are involved in ion transport. The canal thus enclosed between the scala tympani and the scala vestibuli is the cochlear duct (see [Fig. 38.9B](#)). It is triangular in cross-section throughout the length of the cochlea. The closed upper end, the lagena, is attached to the cupula. The lower end of the duct turns medially, narrowing into the ductus reuniens, and connects with the saccule (see [Fig. 38.1C](#)).

The organ of Corti, the sensory epithelium of the cochlea, sits on the basilar membrane. The apices of the sensory hair cells and the supporting cells it contains are joined by tight junctions to form the reticular lamina. The diffusion barriers that line the cochlear duct ensure that the apices of the sensory hair cells are bathed in endolymph, whereas their lateral and basal regions are bathed in perilymph.

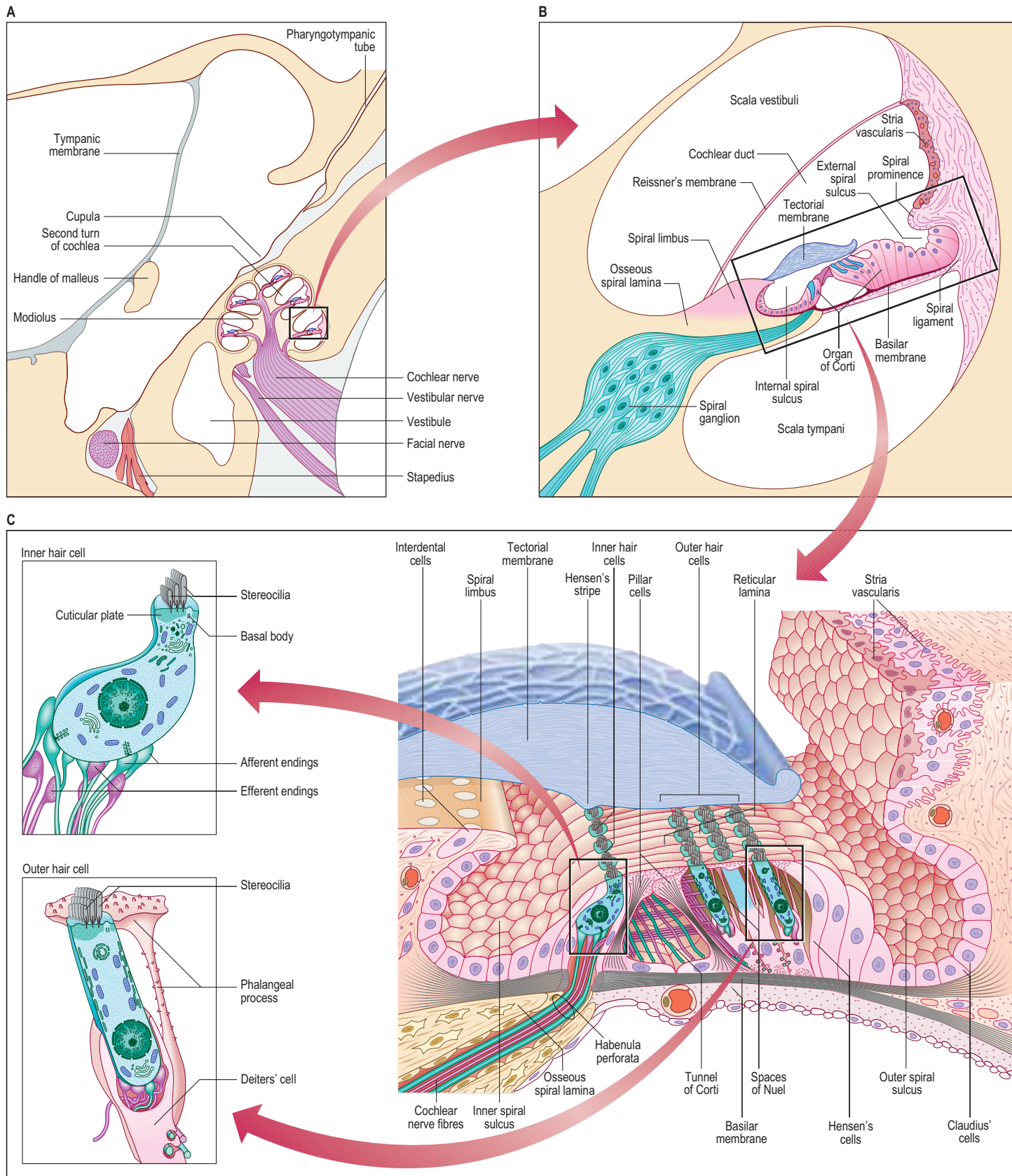


Fig. 38.9 The cochlea. **A**, A horizontal section through the left temporal bone showing the position of the cochlea with respect to the tympanic cavity. **B**, A section through the second turn of the cochlea seen in **A**. The modiolus is to the left. **C**, The structure of the cochlear organ of Corti and stria vascularis, showing the arrangement of the various types of cell and their overall innervation. The organization of the inner and outer hair cells and their synaptic connections are also depicted. **A** and **B** are drawn from histological sections.

The stria vascularis lies on the outer wall of the cochlear duct, above the spiral eminence (see Fig. 38.9B and C). It has a special stratified epithelium containing a dense intraepithelial capillary plexus and three cell types: superficial marginal, dark or chromophil cells; intermediate light, or chromophobe cells; and basal cells. The endolymphatic surface consists only of the apices of marginal cells. The intermediate and basal cells lie deeper within the stria and send cytoplasmic processes towards the surface, between the deeper parts of the marginal cells. The long

descending cytoplasmic processes of the marginal dark cells and the ascending processes of the intermediate and basal cells envelop the intraepithelial capillaries. The stria vascularis is involved in ion transport and helps to produce the unusual ionic composition of endolymph. It is the source of the large positive endocochlear electrical potential, maintenance of which is directly dependent on adequate oxygenation of the epithelial cells, provided by the intraepithelial capillary plexus.

The spiral ligament is lateral to the stria vascularis. It is composed of thick collagenous fibres interspersed with fibrocytic cells of several different types, and root cells that send large processes into the ligament from the region of the basilar crest. The fibrocytes and the root cells express connexins and have gap junctional systems ([Jagger et al 2010](#)) that allow transcellular transport of ions, although the two types of cell are not directly coupled. It is thought that potassium recycling from perilymph to endolymph is mediated partly through these gap junctional networks emanating from the Deiters' cells underlying the sensory hair cells. Late-onset hearing loss (DFN3) is thought to be the result of fibrocyte degeneration ([Minowa et al 1999](#)).

The osseous spiral lamina consists of two plates of bone between which are canals for the cochlear nerve fibres. On the upper plate, the periosteum is thickened to form the spiral limbus (*limbus laminae spiralis*) (see Fig. 38.9B). It ends externally in the internal spiral sulcus, which in section is shaped like a C. Its upper part, the overhanging limbic edge, is the vestibular labium, and the lower tapering part is the tympanic labium, which is perforated by small holes (the *habenula perforata*) for branches of the cochlear nerve. The upper surface of the vestibular labium is crossed at right angles by furrows, separated by numerous elevations – the auditory teeth (*dentes acustici*). The limbus is covered by a layer that appears superficially to be squamous epithelium; however, only the cells over the ‘teeth’ are flat, and those in the furrows are flask-shaped interdental cells. The epithelium is continuous with the epithelium in the internal spiral sulcus and on the inferior surface of Reissner’s membrane. During development, the interdental cells secrete some of the material that forms the tectorial membrane. The limbus contains fibrocytic cells interspersed with extracellular matrix.

Basilar membrane

The basilar membrane stretches from the tympanic lip of the osseous spiral lamina to the basilar crest of the spiral ligament (see Fig. 38.9B and C). It consists of two zones. The thin *zona arcuata* stretches from the spiral limbus to the bases of the outer pillar cells and supports the organ of Corti. It is composed of compact bundles of small (8–10 nm diameter) collagenous filaments, mainly radial in orientation. The outer, thicker *zona pectinata* starts beneath the bases of the outer pillar cells and is attached to the *crista basilaris*. The basilar membrane is trilaminar in the *zona pectinata*, but the upper and lower layers fuse at its attachment to the *crista basilaris*. The length of the basilar membrane is 35 mm; its width increases from 0.21 mm basally to 0.36 mm at its apex, accompanied by corresponding narrowing of the osseous spiral lamina and a decrease in the thickness of the basilar crest. The lower or tympanic surface of the basilar membrane is covered by a layer of vascular connective tissue and elongated perilymphatic cells. One vessel, the spiral vessel (*vas spirale*), is larger; it lies immediately below the tunnel of Corti.

Organ of Corti

The organ of Corti consists of a series of epithelial structures that lie on the *zona arcuata* of the basilar membrane (see Fig. 38.9B and C). The more central of these structures are two rows of cells: the internal (inner) and external (outer) pillar cells. The bases of the pillar cells are expanded, and rest contiguously on the basilar membrane, but their rod-like cell bodies are widely separated. The two rows incline towards each other and come into contact again at the heads of the pillars, enclosing between them and the basilar membrane the tunnel of Corti, which has a triangular cross-section (see Fig. 38.9C). Internal to the inner pillar cells is a single row of inner hair cells. External to the outer pillar cells are three or four rows of outer hair cells. The bases of the outer hair cells are cupped by supporting cells called outer phalangeal (Deiters’) cells, except for a gap where cochlear axons synapse with them. The apical ends of the hair cells and apical processes of the supporting cells form a regular mosaic called the reticular lamina, which is covered by the tectorial membrane, a gel-like structure projecting from the spiral limbus. The reticular lamina is impervious to ions and thus maintains the electrochemical gradient between the fluids surrounding the apices and the basolateral membranes of the sensory hair cells. A narrow gap separates the tectorial membrane from the reticular lamina except where the apical stereocilia of the outer hair cells project to make contact with it.

In addition to the tunnel of Corti, other intercommunicating spaces, the spaces of Nuel, surround the outer hair cells. This entire intercommunicating complex of spaces of Nuel and tunnel of Corti is filled with perilymph, which diffuses through the matrix of the basilar membrane. The fluid in these spaces is also sometimes called the cortilymph; it is possible that minor alterations in perilymphatic composition occur within it because it is exposed to the activities of synaptic endings and specialized excitable cells.

Each pillar cell has a base or crus, an elongated scapus (rod) and an upper end or caput (head) (see Fig. 38.9C); each crus and caput are in contact, but the scapi are separated by the tunnel of Corti. Electron microscopy shows many microtubules, 30 nm in diameter, arranged in linked parallel bundles of 2000 or more in the scapus, originating in the crus and diverging above the scapus to terminate in the head region. The nucleus is situated in the foot-like expansion resting on the basal lamina.

There are almost 6000 inner pillar cells. Their bases rest on the basilar membrane near the tympanic lip of the internal spiral sulcus, and their bodies form an angle of approximately 60° with the basilar membrane. Their heads resemble the proximal end of the ulna, with deep concavities for the heads of the outer pillar cells, which they overhang to form the top of the tunnel of Corti. There are almost 4000 outer pillar cells. They are longer and more oblique than the inner pillar cells, and form an angle of approximately 40° with the basilar membrane. Their heads fit into the concavities on the heads of the inner pillar cells and project externally between the first row of outer hair cells as thin processes that contact the processes of the Deiters’ cells. The distances between the bases of the inner and outer pillar cells increase from the cochlear base to its apex, whereas the angles they make with the basilar membrane diminish.

Cochlear hair cells are the sensory transducers of the cochlea; collectively, they detect the amplitude and frequency of the sound waves that enter the cochlea. All cochlear hair cells have a common pattern of organization. They are elongated cells with a group of modified apical microvilli or stereocilia (which contain parallel arrays of actin filaments), similar to those of vestibular hair cells, and a group of synaptic contacts with cochlear nerve fibres at their rounded bases (Fig. 38.10). The inner hair cells form a single row along the inner edge of

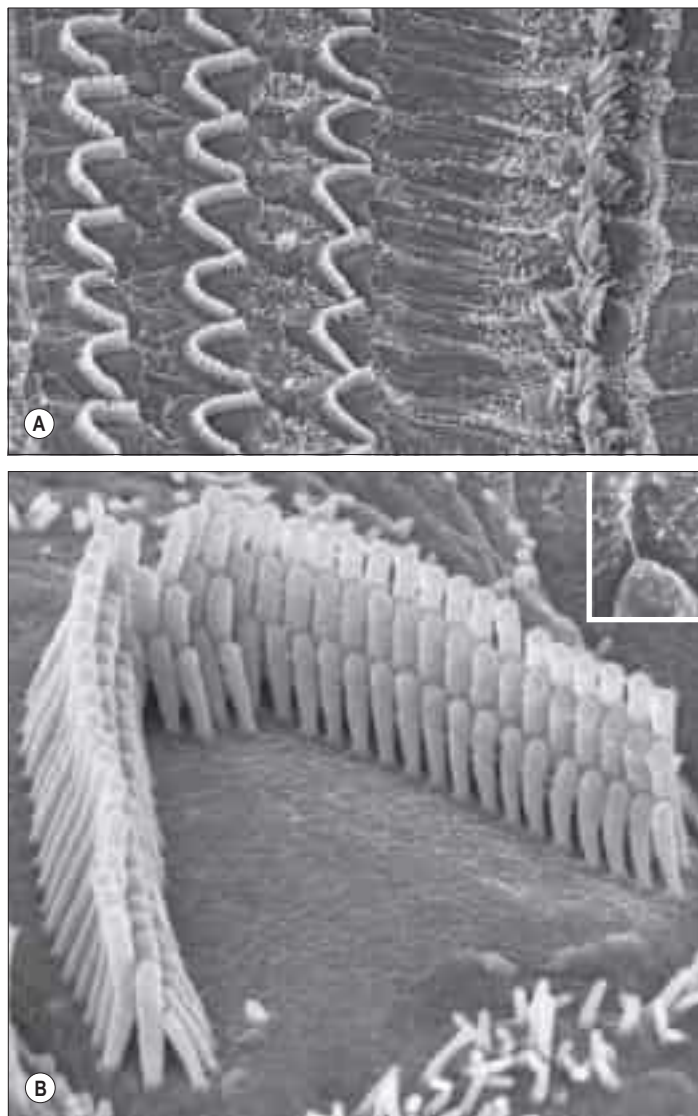


Fig. 38.10 Scanning electron micrographs of the surface of the organ of Corti (guinea pig). **A**, The reticular lamina. Three rows of V-shaped stereociliary bundles can be seen protruding from the apices of the outer hair cells. They are separated from the single row of inner hair cells (which have relatively linear stereociliary bundles) by the apices of the inner pillar cells. **B**, The stereociliary bundle of one outer hair cell, showing three rows of stereocilia increasing in height; deflection of the stereocilia in the direction of the tallest row results in depolarization of the hair cell. Microvilli can be seen on the surface of Deiters’ cells (front right). The inset shows a tip link connecting a short stereocilium tip to the side of the tall stereocilium behind. (Courtesy of DN Furness, The Institute of Science and Technology in Medicine, School of Life Sciences, Keele University.)

inner pillar cells (and the spiral tunnel), whereas the outer hair cells are arranged in three or, in some regions of the human cochlea, in four or even five rows, interspersed with supporting cells (see Fig. 38.10A). These two groups have distinctive roles in sound reception; the differences in their detailed structure reflect this functional divergence. There are 3500 inner hair cells and 12,000 outer hair cells. The two sets of hair cells lean towards each other apically at about the same angles as the neighbouring inner and outer pillar cells. The geometric arrangement of these cells is very precise, and this pattern is closely related to the sensory performance of the cochlea.

The inner hair cells are pear-shaped and slightly curved; the narrower end is directed towards the surface of the organ of Corti and the wider basal end is positioned some distance above the inner end of the basilar membrane (see Fig. 38.9C). The inner hair cells are surrounded by inner border cells and by inner phalangeal cells, which are attached externally to the heads of the inner pillar cells. The flat apical surface of each inner hair cell is elliptical when viewed from above, its long axis directed in the direction of the row of hair cells (see Fig. 38.10A). The breadth of the apex exceeds that of the inner pillar cells so that each inner hair cell is related to more than one inner pillar cell. The apex bears 50–60 stereocilia, arranged in several ranks of progressively ascending height, the tallest on the stria side. The tips of the shorter rows are connected diagonally to the sides of the adjacent taller stereocilia by thin filaments called tip links; each stereocilium is also connected to all its neighbours by a variety of lateral links. The height of a stereociliary row varies along the length of the cochlea, being tallest at the apex and shortest at its base. The stereociliary bases insert into a transverse lamina of dense fibrillar material, the cuticular plate, which lies immediately beneath the apical surface of each inner hair cell. The cuticular plate includes a small aperture containing a basal body. During development, a kinocilium containing microtubules is anchored here, a condition that persists in vestibular hair cells.

At its base, each inner hair cell forms 10 or more synaptic contacts with afferent endings, each being marked by a presynaptic structure similar to the ribbon synapses of the retina. Occasionally, an efferent synapse makes direct contact with a hair cell base but these are usually presynaptic to the terminal expansions of afferent endings, rather than to the hair cell itself.

Outer hair cells are long cylindrical cells that are nearly twice as tall as the inner hair cells (see Figs 38.9C, 38.10A). There is a gradation of length: the outermost row is longest in any one cochlear region, and those of the cochlear apex are taller than those of the base. They are surrounded by the apical or phalangeal processes of the Deiters' cells or, on the internal side of the inner row, by the heads of the outer pillar cells. The stereocilia, which may number up to 100 per cell, are arranged in three rows of graded heights; the tallest is on the outer side. The rows are arranged in the form of a V or W, depending on cochlear region, the points of the angles directed externally. The stereocilia are also graded in height according to cochlear region; those of the cochlear base are shortest. Like those of inner hair cells, the stereocilia possess tip links (see Fig. 38.10B) and other filamentous connections with their neighbours, and are inserted at their narrow bases into a cuticular plate. The tallest stereocilia are embedded in shallow impressions on the underside of the tectorial membrane.

The rounded nucleus is positioned near the base of the cell. Below the nucleus are a few ribbon-like synapses associated with afferent endings of the cochlear nerve. The latter are fewer in number and smaller than the cluster of efferent boutons that contact the base of the cell. The neurotransmitter at the afferent synapse in both inner and outer hair cells is glutamate whereas that of the efferent endings is acetylcholine, although other neurotransmitters or neuromodulators have been demonstrated.

Cochlear hair cells respond with phenomenal speed and sensitivity to sound vibrations that cause submicron deflections of their stereociliary hair bundles. Outer hair cells not only detect these vibrations but also generate force to increase auditory sensitivity and frequency discrimination. (See Fettiplace and Hackney (2006) for further details about the sensory and motor functions of auditory hair cells.)

Deiters' or phalangeal cells lie between the rows of outer hair cells. Their expanded bases lie on the basilar membrane and their apical ends partially envelop the bases of the outer hair cells (see Fig. 38.9C; Fig. 38.11). Each has a finger-like (phalangeal) process that extends diagonally upwards between the hair cells to the reticular membrane, where it forms a plate-like expansion that fills the gaps between hair cell apices.

Five or six rows of columnar supporting cells or external limiting cells, such as Hensen's cells and Claudius' cells, lie external to the Deiters' cells (see Fig. 38.9C).

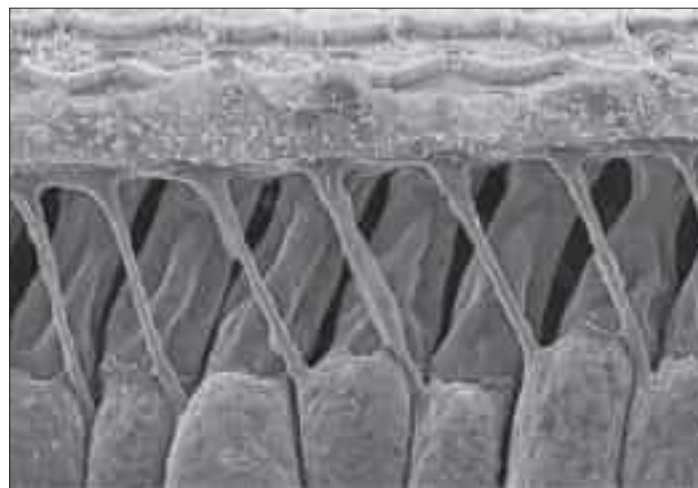


Fig. 38.11 A scanning electron micrograph of a portion of the organ of Corti (guinea pig), dissected to expose the outer row of outer hair cells and their attendant Deiters' cells with narrow phalangeal processes. The stereociliary bundles of two rows of outer hair cells are visible above the reticular lamina. (Courtesy of DN Furness, The Institute of Science and Technology in Medicine, School of Life Sciences, Keele University.)

The apices of the hair cells and supporting cells that form the reticular lamina are linked by tight junctions, adherens junctions and desmosomes. This arrangement is significant for two reasons. The reticular lamina creates a highly impermeable barrier to the passage of ions other than via the mechanotransducer channels in the stereociliary membranes. It also forms a rigid support between the apices of the hair cells, coupling them mechanically to the movements of the underlying basilar membrane, which causes lateral shearing movements between the stereocilia and the overlying tectorial membrane. If there is hair cell loss as a result of trauma such as excessive noise or ototoxic drugs, the supporting cells expand rapidly to fill the gap, disturbing the regular pattern of the reticular lamina (phalangeal scars) but restoring its function.

Hair cell mechanotransduction

Mechanotransduction is central to both hearing and balance, and is a phenomenon that is shared by all the sensory hair cells of vestibular and auditory epithelia. The basic mechanism is the same in all of the different types of hair cell and is dependent on common features of the hair bundle: specifically, the 'staircase' array of stereocilia and the tip links connecting the tips of shorter and adjacent taller stereocilia. Deflections of the bundle in an excitatory (depolarizing) direction cause tension in the tip link, which acts to open non-specific cationic mechanoelectrical transduction channels. These then allow influx of potassium and calcium ions, depolarizing the cell and leading to calcium influx through basolateral voltage-gated calcium channels, which causes release of neurotransmitter. The hair cells also have basolateral potassium channels that allow efflux of potassium into perilymph. The tip link that is central to this process consists of two molecules of cadherin 23 and two molecules of protocadherin 15, forming a strong twisted strand (Hackney and Furness 2013). The importance of this structure is emphasized by the fact that mutations in these cadherins underlie deaf-blindness conditions (Usher syndrome).

The sensitivity of mechanotransduction is greatly increased by the combination of the hair-cell membrane potential (-70 mV) and the endolymphatic potential ($+80$ mV), producing an overall driving potential of up to 150 mV. It is therefore vital to maintain a high potassium concentration, despite its depletion from endolymph during detection of acoustic or balance stimuli.

Tectorial membrane

The tectorial membrane overlies the sulcus spiralis internus and organ of Corti, and is a stiff, gelatinous plate (see Fig. 38.9B and C). It contains collagen types II, V and IX, interspersed with glycoproteins (tectorins), which contribute approximately half of the total protein.

In transverse section, the tectorial membrane has a characteristic shape. The underside is nearly flat and the upper surface is convex, and it is thin on the modiolar side where it is attached to the vestibular labium of the spiral limbus. Its outer part forms a thickened ridge, overhanging the edge of the reticular lamina. The lower surface is relatively smooth, except where the stereocilia of the outer hair cells are

INNER EAR

embedded in the membrane, leaving a pattern of W- or V-shaped indentations: an S-shaped ridge called Hensen's stripe, which projects towards the stereocilia of the inner hair cells. The interdental cells of the spiral limbus are believed to secrete the membrane.

VASCULAR SUPPLY

ARTERIES

The inner ear is principally supplied by the labyrinthine artery. The stylomastoid branch of either the occipital artery or the posterior auricular artery (see Fig. 29.8) also supplies the semicircular canals.

Labyrinthine artery

The labyrinthine artery arises from the basilar artery (see Fig. 19.2), or sometimes from the anterior inferior cerebellar artery. It divides at the bottom of the internal acoustic meatus into cochlear and vestibular branches. The cochlear branch divides into 12–14 twigs, which traverse the canals in the modiolus and are distributed as a capillary plexus to the spiral lamina, basilar membrane, stria vascularis and other cochlear structures. Vestibular arterial branches supply the utricle, saccule and semicircular ducts.

VEINS

Descriptions of the venous drainage of the inner ear are often conflicting, presumably reflecting both normal variation and the existence of connections between the veins that provide the potential for collateral circulation. Veins draining the vestibule and semicircular canals accompany the arteries. They pass towards the utricle to form the vein of the vestibular aqueduct, which empties into either the sigmoid or inferior petrosal sinus. The vein of the cochlear aqueduct (inferior cochlear vein) usually drains into either the inferior petrosal sinus or the superior bulb of the internal jugular vein. It is formed by the union of the common modiolar and vestibulocochlear veins and provides almost all of the venous outflow of the cochlea (Wright and Roland 2013). The common modiolar vein is formed by the union of the anterior and posterior spiral veins near the basal cochlear turn; the vestibulocochlear vein is formed by the confluence of the anterior and posterior vestibular veins and the vein of the round window. When present, a labyrinthine vein drains the apical and middle cochlear coils into either the posterior part of the superior petrosal sinus or the transverse sinus or the inferior petrosal sinus.

For details of the microvasculature of the cochlea of humans and other mammals, see Axelsson (1988).

INNERVATION

VESTIBULOCOCHLEAR NERVE

The vestibulocochlear nerve emerges from the cerebellopontine angle (see Figs 21.4, 21.5). It courses through the posterior cranial fossa in close association with the facial nerve, nervus intermedius and labyrinthine vessels. Together with these structures, it enters the petrous temporal bone via the porus acusticus of the internal acoustic meatus, and divides into an anterior trunk, the cochlear nerve, and a posterior trunk, the vestibular nerve (Fig. 38.12A; see Fig. 37.7). Both contain the centrally directed axons of bipolar neurones, together with a smaller number of efferent fibres that arise from brainstem neurones and terminate on cochlear and vestibular sensory cells. In humans, the intratemporal portion of the vestibulocochlear nerve consists of two histologically distinct portions: a central glial zone adjacent to the brainstem, and a peripheral or non-glial zone (Bridger and Farkashidy 1980). In the glial zone, the axons are supported by central neuroglia, whereas in the non-glial zone they are ensheathed by Schwann cells. The non-glial zone sometimes extends into the cerebellopontine angle medial to the internal acoustic meatus in human vestibulocochlear nerves. During development, a gap of several weeks has been reported between the onset of Schwann cell myelination distally and glial myelination proximally; it has been suggested that the gap may coincide with the time of the final maturation of the organ of Corti. (For further details about the development of the human cochlear nerve, see Ray et al (2005).)

In audiological practice, it is important to distinguish between intratemporal and intracranial lesions. However, this surgical distinc-

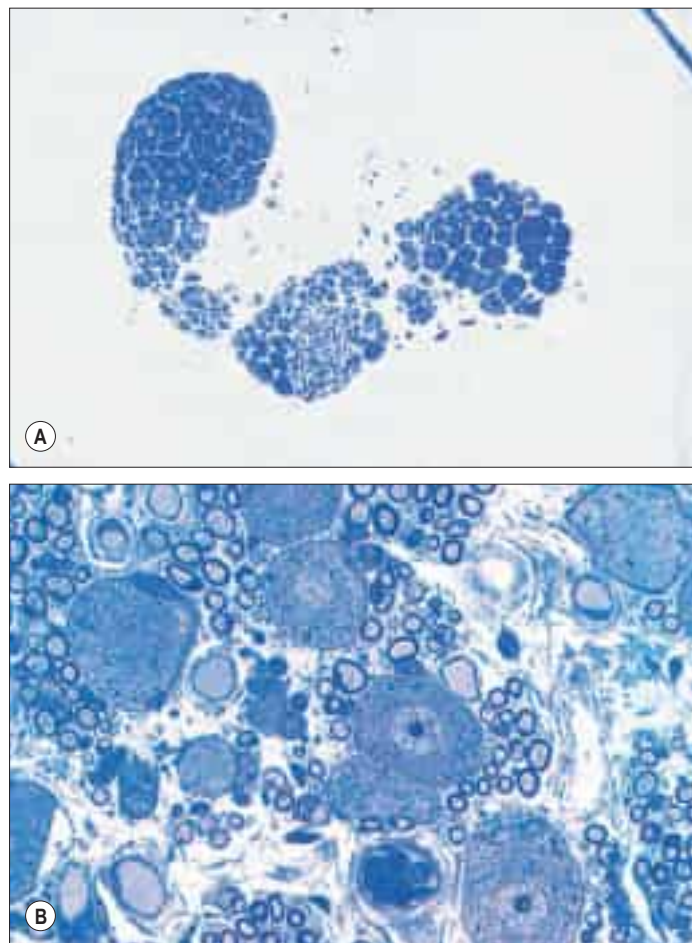


Fig. 38.12 The vestibulocochlear nerve (human). **A**, A transverse section. The cochlear nerve (the comma-shaped profile on the left) abuts the inferior division of the vestibular nerve (right). The singular nerve is a separate fascicle between the superior and inferior divisions of the vestibular nerve. **B**, A portion of a vestibular ganglion, showing neuronal perikarya, myelinated axons and small blood vessels. (Toluidine blue stained resin sections courtesy of H Felix, M Gleeson and L-G Johnsson, ENT Department, University of Zurich and School of Medicine, King's College London.)

tion does not correlate with the precise anatomical description of peripheral and central portions of the auditory and vestibular systems. Clinically, the term 'peripheral auditory lesion' is used to describe lesions peripheral to the spiral ganglion, and the term 'peripheral vestibular disturbance' includes lesions of the vestibular ganglion and the entire vestibular nerve.

Vestibular nerve

The cell bodies of the bipolar neurones that contribute to the vestibular nerve lie in the vestibular ganglion, which is situated in the trunk of the nerve within the lateral end of the internal acoustic meatus (Fig. 38.12B). Their peripheral processes innervate the maculae of the utricle and saccule and the ampullary crests of the semicircular canals (see below). Their axons travel to the central nervous system in the vestibular nerve, which enters the brainstem at the cerebellopontine angle and terminates in the vestibular nuclear complex (see p. 319). Neurones in this complex project to motor nuclei in the brainstem and upper spinal cord, and to the cerebellum and thalamus. Thalamic efferent projections pass to a cortical vestibular area that is probably located near the intraparietal sulcus in area 2 of the primary somatosensory cortex.

Vestibular (Scarpa's) ganglion

The cell bodies of the neurones in the vestibular ganglion vary considerably in size: their circumferences range from 45 to 160 μm (Felix et al 1987). No topographically ordered distribution relating to size has been found. The cell bodies are notable for their abundant granular endoplasmic reticulum, which in places forms Nissl bodies, and prominent Golgi complexes. They are covered by a thin layer of satellite cells and are often arranged in pairs, closely abutting each other so that only a thin layer of endoneurium separates the adjacent coverings of satellite cells. This arrangement has led to speculation that ganglion cells may

affect each other directly by electrotonic spread (ephaptic transmission: see [Felix et al \(1987\)](#)).

Two distinct sympathetic components have been identified in the vestibular ganglion: a perivascular adrenergic system derived from the stellate ganglion, and a blood vessel-independent system derived from the superior cervical ganglion.

Intratemporal vestibular nerve

The peripheral processes of the vestibular ganglion cells are aggregated into definable nerves, each with a specific distribution. The main nerve divides at and within the ganglion into superior and inferior divisions, which are connected by an isthmus. The superior division, the larger of the two, passes through the small holes in the superior vestibular area at the fundus of the internal acoustic meatus (see [Fig. 38.3](#)) and supplies the ampullary crests of the lateral and anterior semicircular canals via the lateral and anterior ampullary nerves, respectively. A secondary branch of the lateral ampullary nerve supplies the macula of the utricle; however, the greater part of the utricular macula is innervated by the utricular nerve, which is a separate branch of the superior division. Another branch of the superior division, supplies part of the saccule.

The inferior division of the vestibular nerve passes through small holes in the inferior vestibular area (see [Fig. 38.3](#)) to supply the remainder of the saccule and the posterior ampullary crest via saccular and singular branches, respectively; the latter passes through the foramen singulare. Occasionally, a very small supplementary or accessory branch innervates the posterior crest; it is probably a vestigial remnant of the crista neglecta, an additional area of sensory epithelium found in some other mammals but seldom in humans.

Afferent and efferent cochlear fibres are also present in the inferior division of the vestibular nerve, but leave at the anastomosis of Oort to join the main cochlear nerve (see review by [Warr \(1992\)](#)). Another anastomosis, the vestibulofacial anastomosis, is situated more centrally between the facial and vestibular nerves, and is the point at which fibres originating in the intermediate nerve pass from the vestibular nerve to the main trunk of the facial nerve.

There are approximately 20,000 fibres in the vestibular nerve, of which 12,000 travel in the superior division and 8000 travel in the inferior division. The distribution of fibre diameters is bimodal, with peaks at 4 µm and 6.5 µm. The smaller fibres go mainly to the type II hair cells and the larger fibres tend to supply the type I hair cells. In addition to the afferents, efferent and autonomic fibres have been identified. Efferent fibres synapse exclusively with the afferent calyceal terminals around type I cells and usually with the afferent boutons on type II cells, although a few are in direct contact with the cell bodies of type II cells. The autonomic fibres do not contact vestibular sensory cells but terminate beneath the sensory epithelia.

Anatomy of balance and posture

The vestibular labyrinths on each side of the head are arranged symmetrically with respect to each other. Vestibular sensory pathways are concerned with perception of the position of the head in space and movement of the head; they also establish important connections for reflex movements that govern the equilibrium of the body and the fixity of gaze.

The vestibular system consists of two otolithic organs – the utricle and the saccule – and three semicircular canals. The otolithic organs detect linear acceleration due to gravitational pull (gravito-inertial acceleration) and the direction of other linear accelerations such as the up-and-down movements of the head that occur in running. They also respond when the head is tilted relative to gravity: so-called pitch (forward and backward tilting) and roll (side-to-side tilting) movements. The semicircular canals detect angular accelerations resulting from rotations of the head or body.

The stereocilia in the apical hair bundles of the mechanosensitive hair cells in each of these organs are embedded in an overlying accessory gel-like structure, the otolithic membrane (in the utricle and the saccule) and the cupula (in the semicircular canals). Their apical surfaces are bathed in endolymph; tight junctional complexes between the apices of the hair cells and their adjacent supporting cells separate the endolymph from the perilymph that bathes their basolateral surfaces. As noted earlier, deflection of the stereocilia (caused by displacements of their overlying accessory membranes by fluid movements in the membranous labyrinth) produces either an increased or a decreased rate of opening of the mechanotransduction channels at their tips, depending on whether they are deflected towards or away from the tallest row, respectively. The change in the membrane potential of the receptor cell is signalled to the brain as a change in the firing frequency

of the vestibular nerve afferents (either an increase or a decrease of the basal resting discharge, depending on the direction of stimulation). The signals are compared centrally with visual and somatosensory signals, which also signal the position of the head in space (for a more detailed account, see [Furness \(2002\)](#)).

Semicircular canals

Angular acceleration and deceleration of the head cause a counterflow of endolymph in the semicircular canals, which deflects the cupula of each crista and bends the stereociliary/kinociliary bundles. When a steady velocity of head movement is reached, the endolymph rapidly adopts the same velocity as the surrounding structures because of friction with the canal walls, so that the cupula and receptor cells return to their resting state. The three semicircular canals are orientated approximately at right angles to each other, which means that all possible directions of acceleration can be detected. Directional sensitivity to head movement is coded by opposing receptor signals: the left and right semicircular canals of each functional pair (e.g. the left and right anterior canals) respond oppositely to any movement of the head that affects them ([Fig. 38.13](#)). Some vestibular neurones receive a bilateral input from vestibular receptors, which means that they can compare the discharge rates of right and left canal afferents, a mechanism that increases the sensitivity of the system.

Maculae

In the maculae, the weight of the otoconial crystals creates a gravitational pull on the otoconial membrane and thus on the stereociliary bundles of the sensory cells that are inserted into its base. Because of this, they are able to detect the static orientation of the head with respect to gravity. They also detect shifts in position according to the extent to which the stereocilia are deflected. The two maculae are set at right angles to each other, and the cells of both are orientated functionally in opposite directions across their striolar boundaries. Movement causes depolarization of the hair cells on one side of the striola and hyperpolarization of cells on the other side; because the striola is curved, small groups of hair cells on the macular epithelium each respond to a specific direction of head tilt or linear acceleration ([Fig. 38.14](#)). Moreover, because the otoconia have a collective inertia/momentum, linear acceleration and deceleration along the anteroposterior axis can be detected by the lag or overshoot of the otoconial membrane with respect to the epithelial surface, and so the saccular macula is able to signal these changes of velocity.

The macular receptors can also be stimulated by low-frequency sound, which sets up vibratory movements in the otoconial membrane, although this appears to require relatively high sound levels. Efferent synapses on the afferent endings of the type I sensory cells and on the bases of type II cells receive inputs from the brainstem that appear to be inhibitory. They serve to reduce the activity of the afferent fibres either indirectly, in the case of the type I cells, or directly, for the type II cells.

Visual reflexes

The vestibular system plays a major role in the control of visual reflexes, which allow the fixation of gaze on an object in spite of movements of the head, and require the coordinated movements of the eye, neck and upper trunk. Constant adjustments of the visual axes are achieved chiefly through the medial longitudinal fasciculus, which connects the vestibular nuclear complex with neurones in the oculomotor, trochlear and abducens nuclei and with upper spinal motor neurones ([Fig. 38.15](#); see [Fig. 41.12](#)), and also by the vestibulospinal tracts. See [page 347](#) and [Fig. 22.24](#) for further information about the vestibulo-ocular reflex.

Abnormal activity of the vestibular input or central connections has various effects on these reflexes, e.g. the production of nystagmus. This can be elicited by the caloric test, a clinical test of vestibular function, by syringing the external acoustic meatus with water above or below body temperature, a procedure that appears to stimulate the cristae of the lateral semicircular canal directly. Spontaneous high activity in the afferent fibres of the vestibular nerve is seen in Ménière's disease, in which those affected experience a range of disturbances, including the sensation of dizziness and nausea, the latter reflecting the vestibular input to the vagal reflex pathway.

Cochlear nerve

Intratemporal cochlear nerve

The cochlear nerve connects the organ of Corti to the cochlear nuclei and related nuclei of the brainstem. The cochlear nerve lies inferior to the facial nerve throughout the internal acoustic meatus (see above). It

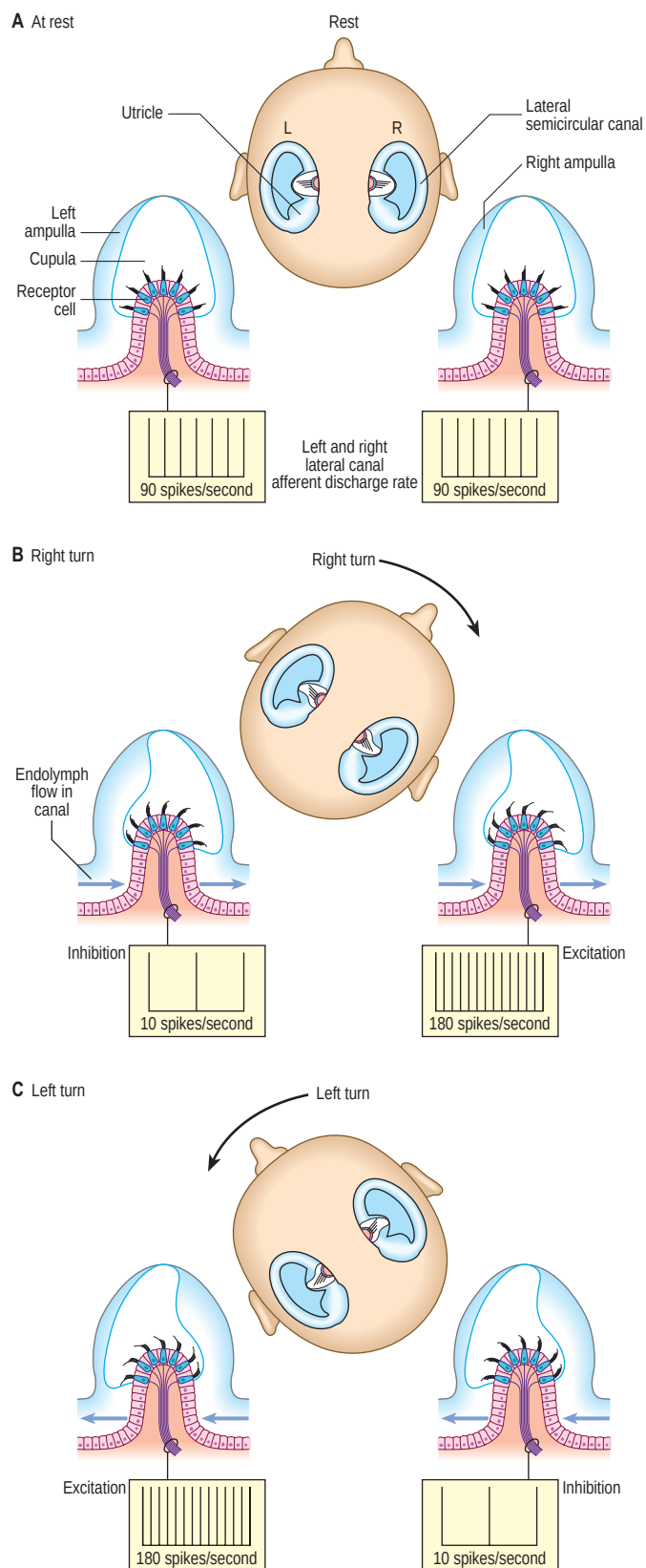


Fig. 38.13 The response of the lateral semicircular canals to head rotation in the horizontal plane. The firing rates of afferents from the left and right lateral canals are equivalent at rest (**A**). However, when the head is turned to the right (**B**) or to the left (**C**), receptor depolarization and afferent fibre excitation occurs on the side to which the head turns; there is inhibition on the contralateral side.

becomes intimately associated with the superior and inferior divisions of the vestibular nerve, which are situated in the posterior compartment of the canal, and leaves the internal acoustic meatus in a common fascicle (see Fig. 38.12A).

There are 30–40,000 nerve fibres in the human cochlear nerve (for review, see Nadol (1988)). Their fibre diameter distribution is unimodal, and ranges from 1 to 11 μm , with a peak at 4–5 μm . Functionally, the nerve contains both afferent and efferent somatic fibres, together

with adrenergic postganglionic sympathetic fibres from the cervical sympathetic system.

Afferent cochlear innervation

The afferent fibres are myelinated axons with bipolar cell bodies that lie in the spiral ganglion in the modiolus (see Fig. 38.9B; Fig. 38.16). There are two types of ganglion cell: most (90–95%) are large type I cells, while the remainder are smaller type II cells (see review by Nadol (1988)). Type I cells contain a prominent spherical nucleus, abundant ribosomes and many mitochondria; in many mammals (although possibly not in humans), they are surrounded by myelin sheaths. In contrast, type II cells are smaller and always unmyelinated, and have a lobulated nucleus. The cytoplasm of type II cells is enriched with neurofilaments but has fewer mitochondria and ribosomes than type I cells.

Basilar fibres Basilar fibres are the peripheral processes of type II ganglion cells and are afferent to the outer hair cells. They take an independent spiral course, turning towards the cochlear apex near the bases of the inner hair cells. They run for a distance of about five pillar cells before turning radially again and crossing the floor of the tunnel of Corti, often diagonally, to form part of the outer spiral bundle.

Outer spiral bundles The afferent fibres of the bundles of the outer spiral group course towards the basal part of the cochlea, continually branching off *en route* to supply several outer hair cells. The outer spiral bundles also contain efferent fibres (see below).

Efferent cochlear fibres

The efferent nerve fibres in the cochlear nerve are derived from the olivocochlear system (see reviews by Warr (1992), Guinan (1996)). Within the modiolus, the efferent fibres form the intraganglionic spiral bundle, which may be one or more discrete groups of fibres situated at the periphery of the spiral ganglion (Fig. 38.17). There are two main groups of olivocochlear efferents: lateral and medial. The lateral efferents come from small neurones in and near the lateral superior olivary nucleus and arise mainly, but not exclusively, ipsilaterally. They are organized into inner spiral fibres that run in the inner spiral bundle before terminating on the afferent axons that supply the inner hair cells. The medial efferents originate from larger neurones in the vicinity of the medial superior olivary nucleus, and the majority arise contralaterally. They are myelinated and cross the tunnel of Corti to synapse with the outer hair cells mainly by direct contact with their bases, although a few synapse with the afferent terminals. The efferent innervation of the outer hair cells decreases along the organ of Corti from cochlear base to apex, and from the first (inner) row to the third. The efferents use acetylcholine, γ -aminobutyric acid (GABA) or both as their neurotransmitter. They may also contain other neurotransmitters and neuromodulators.

Activity of the medial efferents inhibits cochlear responses to sound; the strength of the activity grows slowly with increasing sound level. They are believed to modulate the micromechanics of the cochlea by altering the mechanical responses of the outer hair cells, thus changing their contribution to frequency selectivity and sensitivity. The lateral efferents related to the inner hair cells also respond to sound. They make contacts most commonly with the inner radial afferent fibres rather than the inner hair cell base (see Fig. 38.9C) and appear to modify transmission through the afferents. The cholinergic fibres may excite the radial fibres, while those containing GABA may inhibit them, although their role is less well understood than that of the medial efferents (see review by Guinan (1996)).

Autonomic cochlear innervation

Autonomic nerve endings appear to be entirely sympathetic. Two adrenergic systems have been described within the cochlea: a perivascular plexus derived from the stellate ganglion and a blood vessel-independent system derived from the superior cervical ganglion. Both systems travel with the afferent and efferent cochlear fibres and seem to be restricted to regions away from the organ of Corti. The sympathetic nervous system may cause primary and secondary effects in the cochlea by remotely altering the metabolism of various cell types and by influencing the blood vessels and nerve fibres with which it makes contact.

Anatomy of hearing

Sound waves entering the external ear are converted into electrical signals in the cochlear nerve by the peripheral auditory system (Fig. 38.18). The axons in the cochlear nerve constitute the auditory component of the vestibulocochlear nerve and terminate in the dorsal

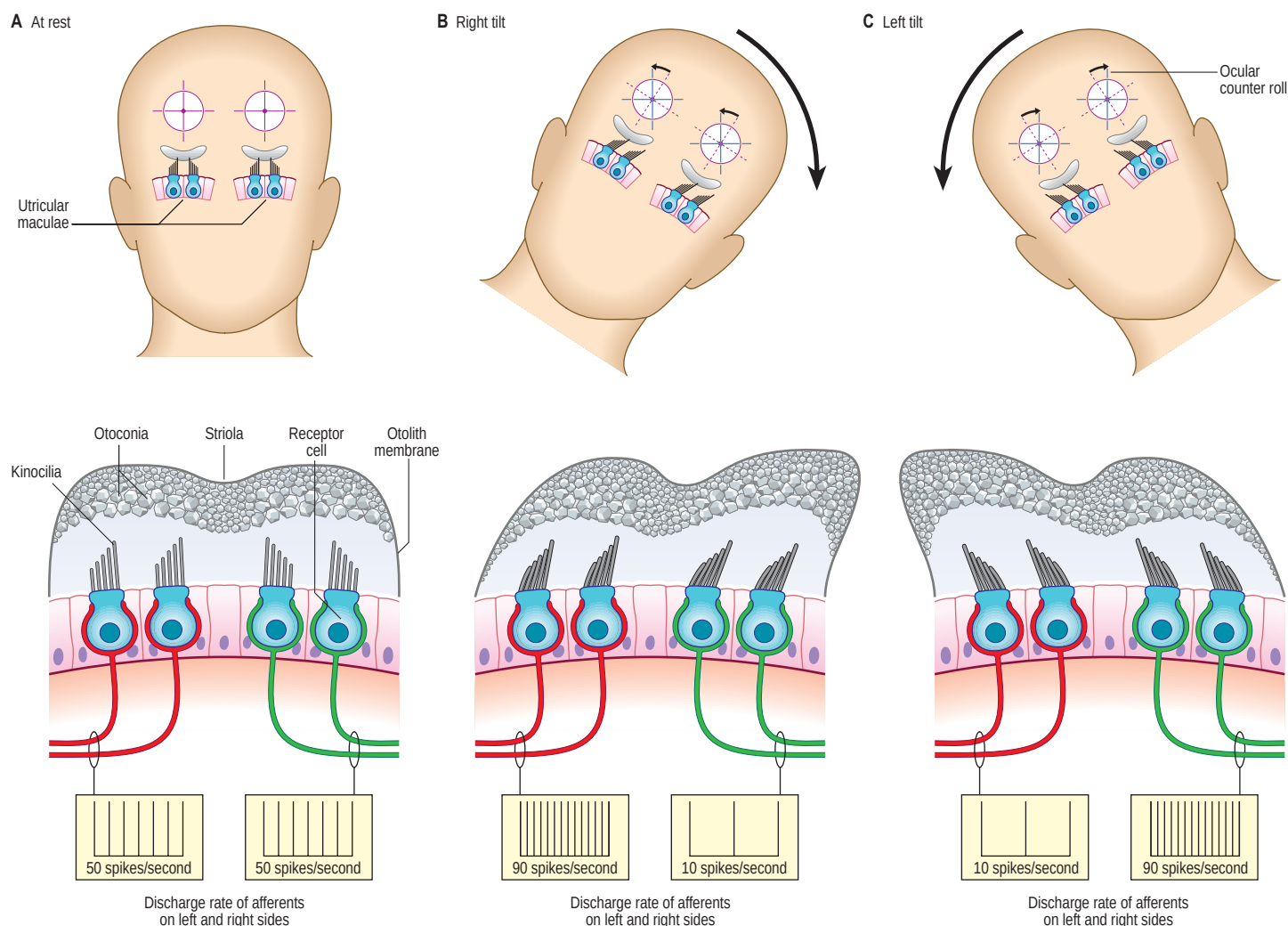


Fig. 38.14 Head tilt is encoded by a macular map of directional space. These diagrams depict the responses of the utricular maculae to head tilt. Firing rates in the vestibular afferents that innervate receptors on either side of the striola (red and green lines) are equivalent when the head is upright (**A**). When the head is tilted to the right (**B**) or to the left (**C**), the stereocilia are deflected by displaced otoconia; hair cells on the upward slope side of the striola increase their firing rate, while those on the downward slope decrease their firing rate.

and ventral cochlear nuclei (see p. 319); onward connections make up the ascending (central) auditory pathway.

Peripheral auditory system

Vibrations in the air column in the external acoustic meatus cause a comparable set of vibrations in the tympanic membrane and auditory ossicles. The chain of ossicles acts as a lever that increases the force per unit area at the round window by 1.2 times while the reduction in size of the round window compared with the tympanic membrane increases the force per unit area of the oscillating surface a further 17 times. This overcomes the inertia of the cochlear fluids and produces in them pressure waves that are conducted almost instantaneously to all parts of the basilar membrane. The latter varies continuously in width, mass and stiffness from the basal to the apical end of the cochlea. Each part of the basilar membrane vibrates, but only the region tuned to a specific frequency will respond maximally to a pure tone entering the ear. A wave of mechanical motion, the travelling wave, is propagated along the basilar membrane to the position where it responds maximally and then dies away again. With increasing frequency, the locus of maximum amplitude moves progressively from the apical to the basal end of the cochlea. The pattern of vibrations in the basilar membrane thus varies with the intensity and frequency of the acoustic waves reaching the perilymph. Because of the arrangement of the hair cells on the basilar membrane, these oscillations generate a largely transverse shearing force between the outer hair cells and the overlying tectorial membrane (in which the apices of the hair cell stereocilia are embedded). This movement depends on the mechanical properties of the entire organ of Corti, including its cytoskeleton, which stiffens this structure. The inner hair cell stereocilia, which probably do not touch the tectorial membrane, although they come very close to it, are likely to be stimulated by local movements of the endolymph. Displacement of the stereociliary bundle of a hair cell activates mechanoelectrical transduc-

tion channels near the tips of its stereocilia, and this allows potassium and calcium ions from the endolymph to enter the hair cell (see earlier and overview by Fettiplace and Hackney (2006)). This induces a depolarizing receptor potential and the release of neurotransmitter on to the cochlear afferents at the base of the cell. In this way, a specific group of auditory axons is activated at the position of maximal basilar membrane vibration.

The mechanical behaviour of the basilar membrane is responsible for a broad discrimination between different frequencies (passive tuning; see overview by Ashmore (2002)), but fine frequency discrimination in the cochlea appears to be related to physiological differences between the hair cells. Individual tuning of hair cells may result from differences in shape, stereociliary length, or possibly variations in the molecular composition of sensory membranes, and may have a role in cochlear amplification (active tuning).

The activity of the outer hair cells appears to play an important part in regulating inner hair-cell sensitivity at specific frequencies. Outer hair cells can change length when stimulated electrically at frequencies of many thousands of cycles per second. The rapidity of these changes in length indicates a novel type of motile mechanism, which is believed to depend on conformational changes in prestin, a protein that displays piezoelectricity and is located in the plasma membrane of the cells (Fettiplace and Hackney 2006, Hudspeth 2014) (Fig. 38.19). When the membrane potential of the outer hair cells changes, they generate forces along their axes. When the mechanoelectrical transduction channels open, they are thought to oppose the viscous forces that tend to damp down the vibration of the cochlear partition, and adjust the mechanics of the organ of Corti on a cycle-by-cycle basis. Alternatively, they may alter the mechanics of the partition more slowly under the influence of the efferent pathway. At a particular frequency, an increase in the intensity of stimulus is signalled by an increase in the rate of discharge in individual cochlear axons. At greater

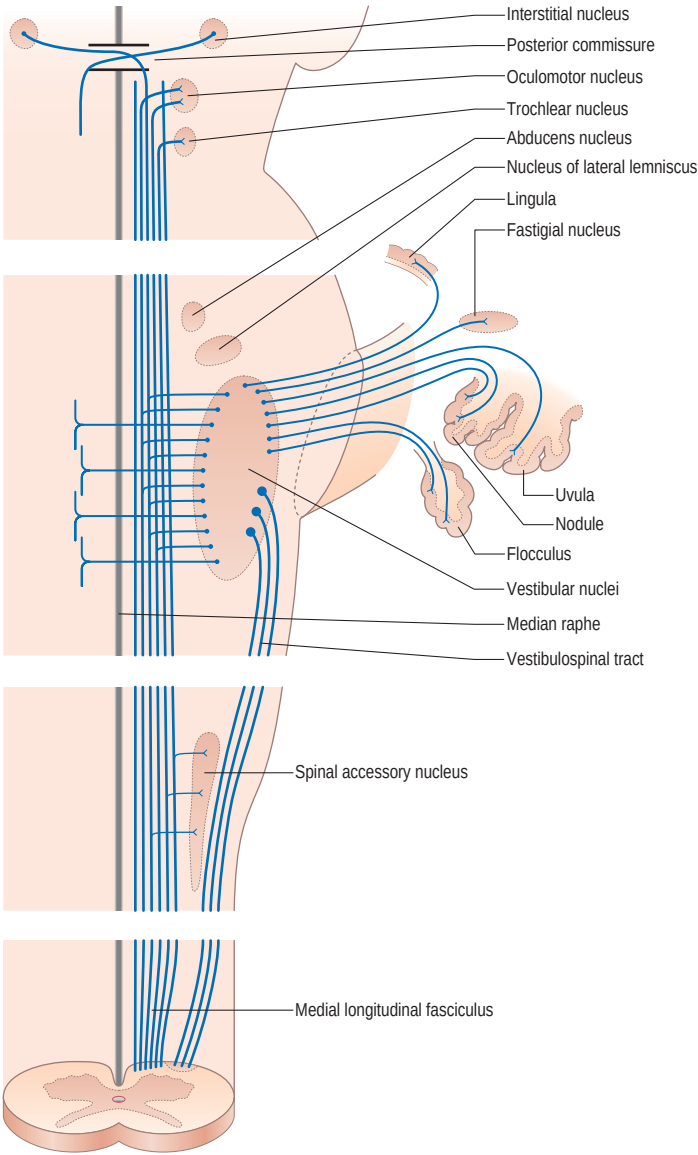


Fig. 38.15 Some of the fibre components of the medial longitudinal fasciculus.



Fig. 38.16 A transmission electron micrograph showing several type II ganglion cells and axons in a human spiral ganglion. Note the absence of myelin from the surrounding sheaths of the ganglion cells. (Courtesy of H Felix, M Gleeson and L-G Johnsson, ENT Department, University of Zurich and GKT School of Medicine, London.)

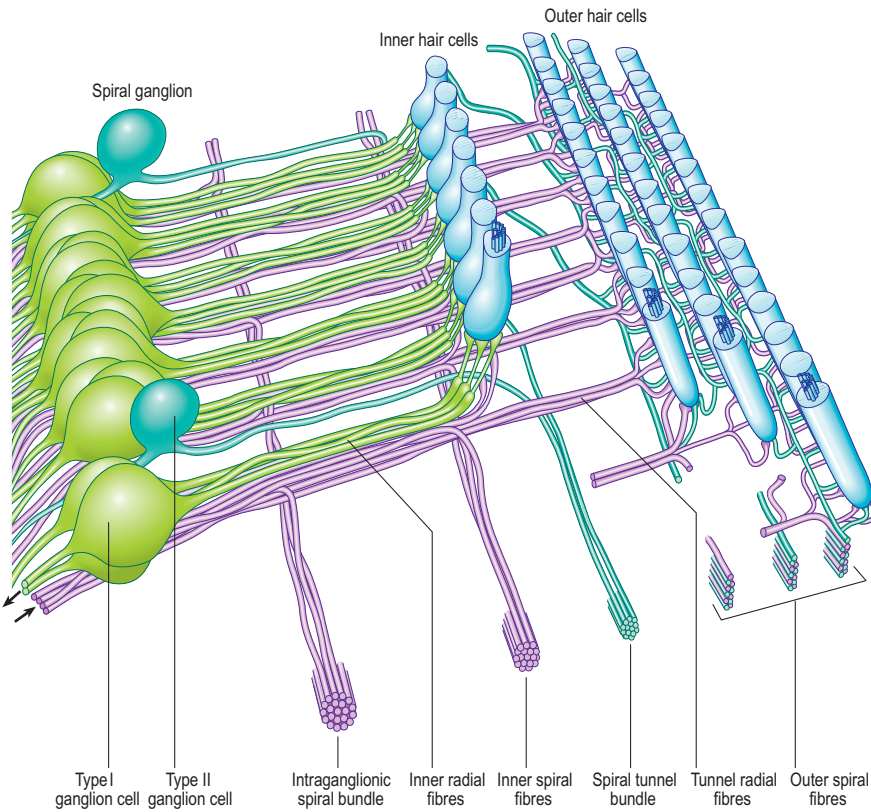


Fig. 38.17 A simplified view of the innervation of the organ of Corti. Note the contrast between the convergent afferent innervation of the inner hair cells (approximately 10 fibres to each cell) and the divergent supply of the outer hair cells (1 afferent fibre to 10 cells).

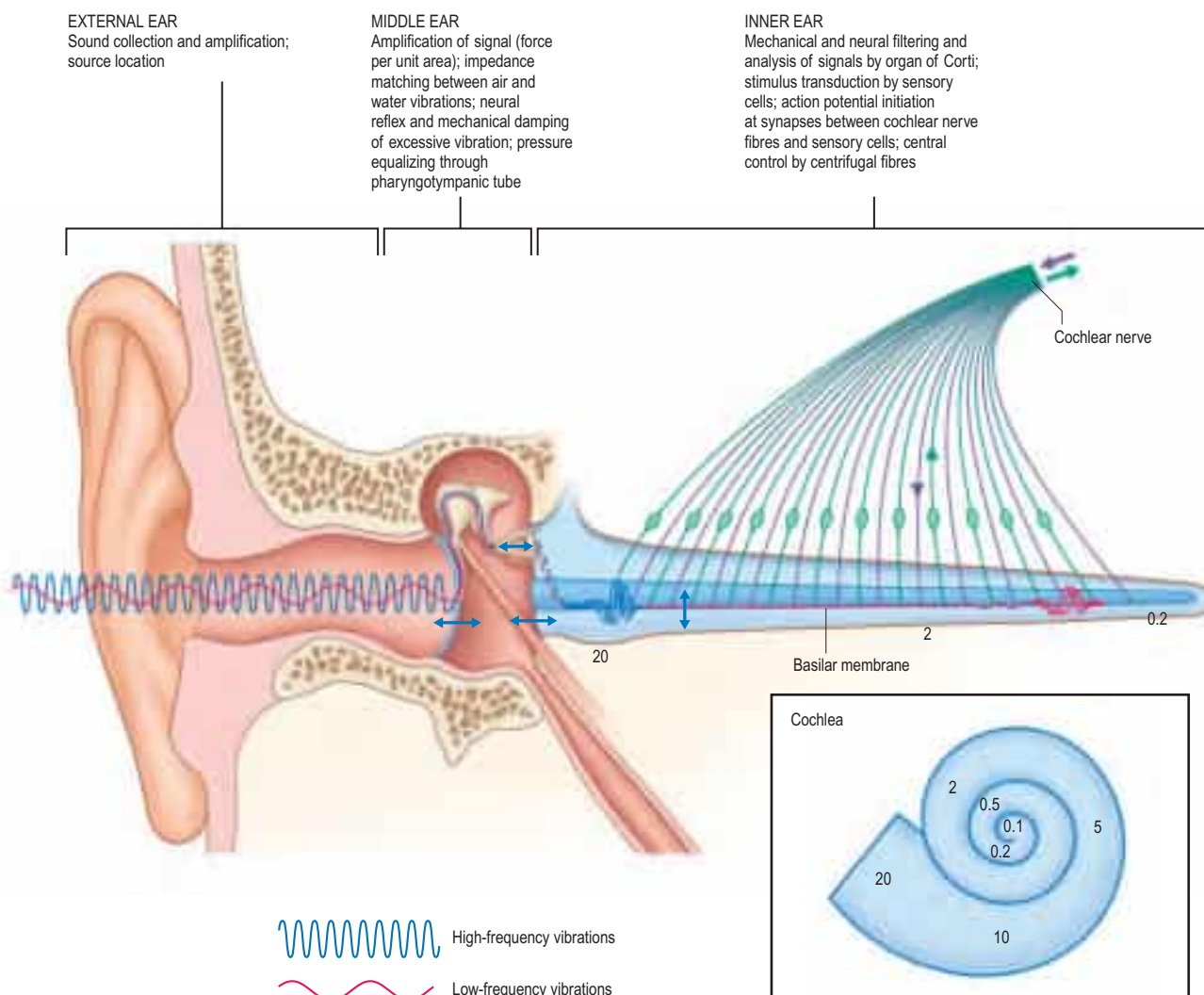


Fig. 38.18 The principal activities of the peripheral auditory apparatus. For clarity, the cochlea is depicted as though it had been uncoiled, but it is normally coiled as in the inset. Different sound frequencies differentially excite different regions of the cochlea, the specific locations being given in kHz from 0.1 to 20 kHz in humans. Note that the frequency map is logarithmic, so that each decade occupies an equivalent distance on the basilar membrane. The components are drawn roughly to scale for the human ear, in which the cochlea is 35 mm in length. The points of maximal stimulation of the basilar membrane by high-frequency and low-frequency vibrations, together with their transmission pathways through the external and middle ear, are also indicated.

intensities, it is signalled by the number of activated cochlear axons (recruitment).

The respective roles of the two groups of hair cells have been much debated, particularly since differences in their innervation and physiological behaviour have become apparent. Because of their rich afferent supply, inner hair cells are believed to be the major source of auditory signals in the cochlear nerve. Some evidence for this view is based on the finding that animals treated with antibiotics that are specifically toxic to outer hair cells are still able to hear, but their sensitivity and frequency discrimination is impaired.

Some electrical responses of the cochlea can be recorded with extracellular electrodes. The most significant is the endolymphatic potential, a steady potential recordable between the cochlear duct and the scala tympani, which is caused by the different ionic compositions of their fluids. As the resting potential of hair cells is approximately 70 mV (negative inside) and the endolymphatic potential is positive in the cochlear duct, the total transmembrane potential across the apices of hair cells is 150 mV. This is a greater resting potential than is found anywhere else in the body, and provides the driving force for mechanotransduction and for the cochlear amplifier.

Under stimulation by sound, a rapid oscillatory cochlear microphonic potential can be recorded. It matches the frequency of the stimulus and movements of the basilar membrane precisely, and appears to depend on fluctuations in the conductance of hair cell membranes, probably of the outer hair cells. At the same time, an extracellular summing potential develops, a steady direct current shift related to the (intracellular) receptor potentials of the hair cells. Cochlear nerve fibres then begin to respond with action potentials that are also recordable from the cochlea. Intracellular recording of auditory responses from inner hair cells has confirmed that these cells resemble other

receptors; their steady receptor potentials are related in size to the amplitude of the acoustic stimulus. At the same time, afferent axons are stimulated by synaptic action at the bases of the inner hair cells. They fire more rapidly as the vibration of the basilar membrane increases in amplitude, up to a threshold that depends on the sensitivity of the specific nerve fibre involved. Each inner hair cell is contacted by axons with response thresholds that range from 0 decibels sound pressure level (dB SPL), the approximate threshold of human hearing, to those that respond to intensities in excess of 100 dB SPL; the loudest sound tolerable is around 120 dB SPL. Each axon responds most sensitively to the frequency represented by its particular cochlear location, its characteristic frequency (see Fig. 38.18).

Central auditory pathway

The primary afferents of the auditory pathway arise from cell bodies in the spiral ganglion of the cochlea. The axons travel in the vestibulocochlear nerve, which enters the brainstem at the cerebellopontine angle. Afferent fibres bifurcate, and terminate in the dorsal and ventral cochlear nuclei (Fig. 38.20, see p. 319). The dorsal cochlear nucleus projects via the dorsal acoustic stria to the contralateral inferior colliculus. The ventral cochlear nucleus projects via the trapezoid body or the intermediate acoustic stria to relay centres in either the superior olivary complex, the nuclei of the lateral lemniscus, or the inferior colliculus. The superior olivary complex is dominated by the medial superior olivary nucleus, which receives direct input from the ventral cochlear nucleus on both sides, and is involved in localization of sound by measuring the time difference between afferent impulses arriving from the two ears.

The inferior colliculus consists of a central nucleus and two cortical areas. The dorsal cortex lies dorsomedially, and the external cortex lies

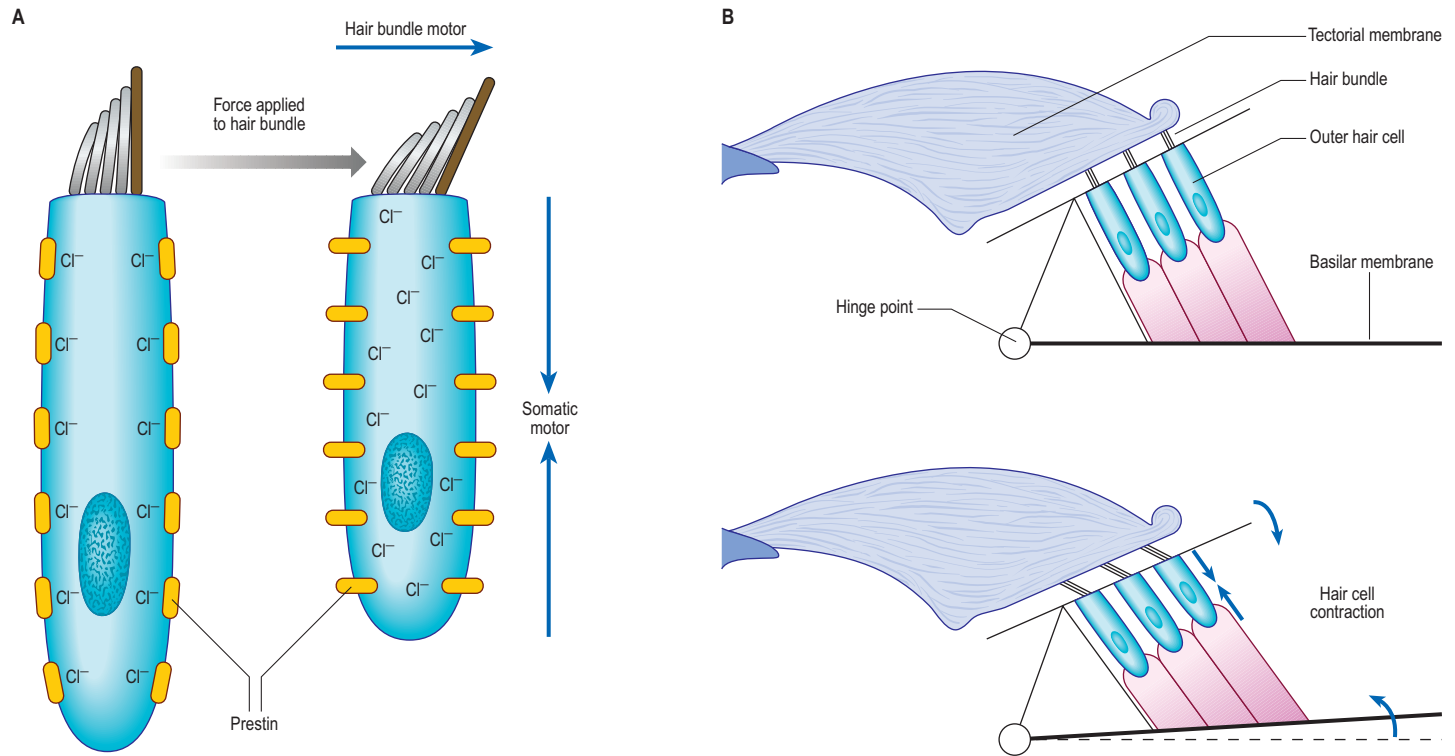


Fig. 38.19 The putative motors of outer hair cells. Outer hair cells can generate force, mechanically boosting sound-induced vibrations of the hair bundle and augmenting frequency tuning. Two mechanisms have been advanced to explain this cochlear amplifier: the somatic motor and the hair bundle motor. **A**, In the resting state, Cl^- ions are bound to prestin molecules in the lateral membrane of the hair cell. When force is applied to the hair bundle, the cell is depolarized, the Cl^- ions dissociate and the prestin changes conformation, reducing its area in the plane of the membrane and shortening the hair cell body (the somatic motor). Adaptation of mechanoelectrical transduction (MET) channels, which are activated by bending of the stereocilia at their tapered base, also causes the hair bundle to produce extra force in the direction of the stimulus (the hair bundle motor). The amplitudes of the hair bundle movements have been exaggerated to illustrate the concept. **B**, The effects of the somatic motor (blue arrows) on the organ of Corti mechanics, which leads to downward motion of the reticular lamina (the upper surface of the organ of Corti) and a negative deflection of the hair bundle. This is a negative feedback pathway, as a positive deflection of the hair bundle causes outer hair cell depolarization, cell contraction and opposing motion of the bundle (see Fettiplace and Hackney (2006)).

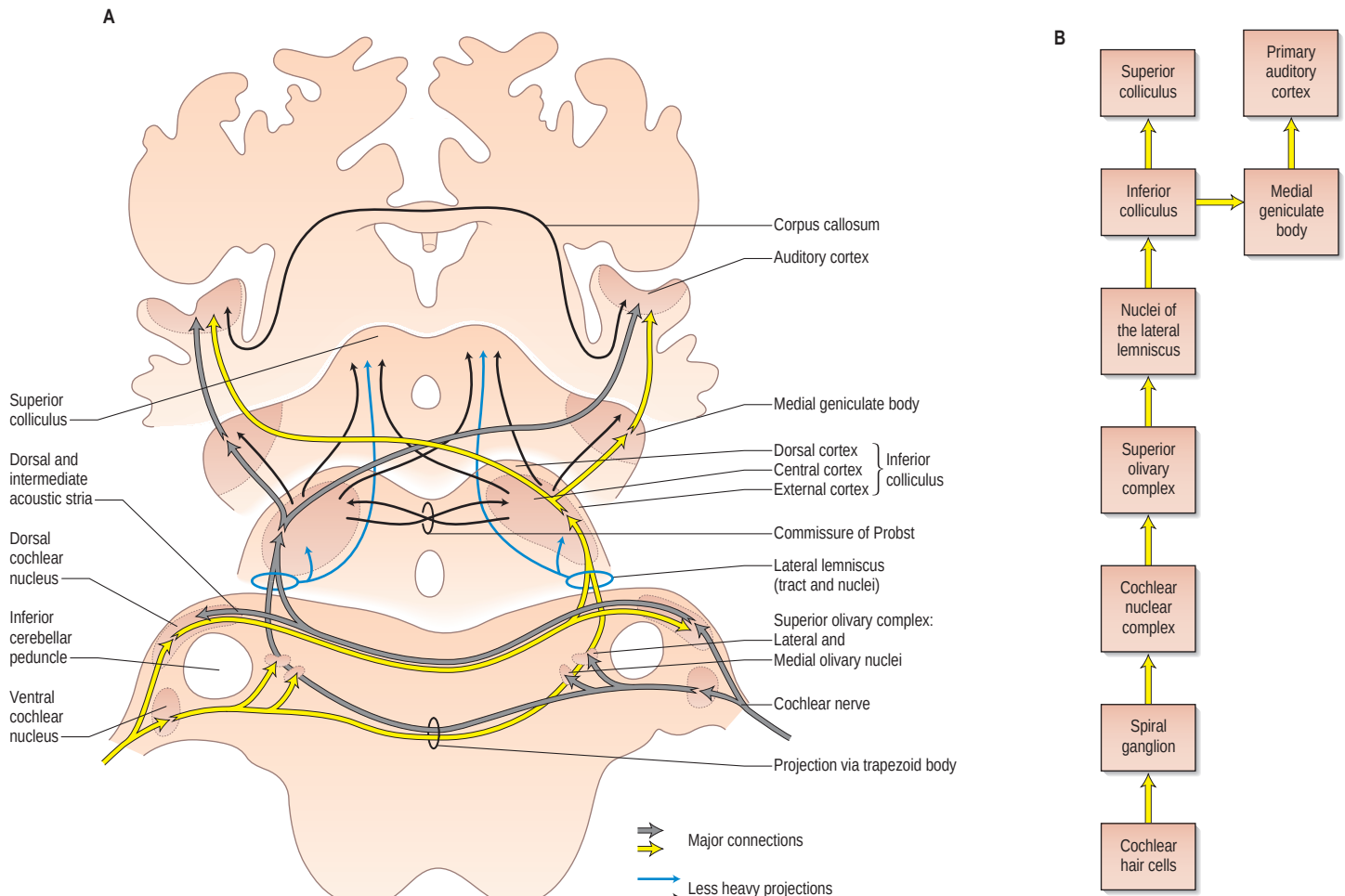


Fig. 38.20 The main features of the human ascending auditory pathway. **A**, A series of sections showing that ipsilateral and commissural connections occur at most levels in this system. The major connections are shown by the thicker arrows; thinner arrows denote less heavy projections. **B**, The main stations of the auditory pathway.

ventromedially. Secondary and tertiary fibres ascend in the lateral lemniscus. They converge in the central nucleus, which projects to the ventral division of the medial geniculate body of the thalamus. The external cortex receives both auditory and somatosensory input. It projects to the medial division of the medial geniculate body and, together with the central nucleus, also projects to olivocochlear cells in the superior olivary complex and to cells in the cochlear nuclei. The dorsal cortex receives input from the auditory cortex and projects to the dorsal division of the medial geniculate body. Connections also run from the nucleus of the lateral lemniscus to the deep part of the superior colliculus, to coordinate auditory and visual responses.

The ascending auditory pathway crosses the midline at several points both below and at the level of the inferior colliculus. However, the input to the central nucleus of the inferior colliculus and higher centres has a clear contralateral dominance; during the initial stages of cortical auditory processing, both hemispheres respond most strongly to the contralateral ear. The medial geniculate body is connected reciprocally to the primary auditory cortex, which lies in the posterior half of the superior temporal gyrus and also dives into the lateral sulcus as the transverse temporal gyri (Heschl's gyri). Secondary areas of the auditory cortex are located in an adjacent belt region, and other regions of auditory association cortex have been described in a parabelt region beyond the secondary cortex.

The corpus callosum, particularly the posterior third of the body, contains auditory interhemispheric fibres that originate from the primary and second auditory cortices. Asymmetries of minicolumn number in primary and association auditory regions have been correlated with axonal fibre numbers in the subregions of the corpus callosum through which they project (Chance et al 2006).

The presence of tonotopic gradients in the primary auditory cortex is well established in animals and in humans. Hemispheric differences for frequency selectivity (i.e. the ability of the cochlea to separate the acoustic frequencies along its length like an acoustic prism) and tonotopic organization have been reported, e.g. the right hemisphere appears to be most responsive to acoustic sound features such as pitch, whereas the left hemisphere seems to be more involved in processing temporal dynamics such as the phonological aspects of speech. Morphological asymmetries favouring the left hemisphere in the planum temporale and Heschl's gyri have been correlated with left hemispheric dominance for language functions but a direct link between structure and function has not been clearly established; studies often show relative rather than absolute differences in hemispheric specialization for particular attributes.

The transformation of the physical characteristics of sound into 'auditory objects' is thought to occur in the transition from primary to secondary auditory cortex. (For a critical perspective on auditory objects, see Griffiths and Warren (2004).)

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Deafness

Hearing impairment is the most common disabling sensory defect in humans. Two causes of deafness are usually distinguished: conductive hearing loss and sensorineural hearing loss.

Conductive hearing loss may result from trauma to the external or middle ears, blockage of the external acoustic meatus, or disruption of the tympanic membrane (e.g. by intense sounds or extreme pressure changes). It may also result from acute or chronic infection associated with a perforation of the tympanic membrane, as in acute otitis media. The ossicular chain may be disrupted by trauma, avascular necrosis of the long process of the incus following a middle ear infection or cholesteatoma. The most common cause of a conductive hearing loss is a middle ear effusion (glue ear), in which fluid that has replaced the normally air-filled middle ear impedes the movements of the tympanic membrane. Other less common causes include tympanosclerosis, in which calcium plaques are formed in the tympanic membrane and middle ear mucosa that stiffen the tympanic membrane and limit the mobility of the ossicular chain. Similarly, new bone laid down around the oval window and footplate of the stapes, otosclerosis, impedes stapes mobility and imparts a conductive hearing loss.

Sensorineural hearing loss is the most prevalent form of hearing impairment. It refers commonly to loss or damage of the sensory hair cells or their innervation, but can also result from lateral wall degeneration and loss of the endolymphatic potential. The hair cells are particularly vulnerable to mechanical trauma produced by high-intensity noise and to changes in their physiological environment caused by infection or hypoxia. These can lead to rapid degenerative processes that result in hair cell loss, often by apoptosis, and produce either hearing loss or vestibular dysfunction. These changes can be induced by drugs such as the aminoglycoside antibiotics, some diuretics and certain anticancer drugs. A decrease in cochlear sensitivity, presbycusis, almost invariably occurs with age; hair cells at the high frequency end of the cochlea tend to be lost first. At least 60% of hearing loss may have a genetic basis, a significant proportion may be non-syndromic, and most of these genes are inherited in an autosomal recessive mode (see Ch. 39).

Ménière's disease is a distressing disorder of the inner ear characterized by episodes of hearing loss, tinnitus and vertigo. Histological examination of an affected ear reveals endolymphatic hydrops (swelling of the endolymphatic spaces), suggesting poor drainage of the endolymph via the endolymphatic sac.

Surgical approaches to the inner ear



Available with the *Gray's Anatomy* e-book

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A review of inner ear fluids (perilymph and endolymph), how they are produced and their possible functional significance.

The inner ear may be approached surgically in a number of ways. Many drugs, e.g. gentamicin and steroids, are now injected through the intact tympanic membrane, where they diffuse through the round window membrane to access the inner ear. The promontory that overlies the basal turn of the cochlea and the oval window may be opened through a tympanotomy (per-meatal elevation of the tympanic membrane). The semicircular canals are generally accessed by a transmastoid approach. In days gone by, such an approach would be used to fenestrate the lateral canal in the management of otosclerosis. The advent of stapes prostheses and stapedectomy rendered this operation obsolete. However, the transmastoid approach is the workhorse for much middle and inner ear surgery because it gives a wide exposure of all the vital structures. For patients with profound hearing loss, enlarging the mastoid aditus inferiorly (posterior tympanotomy) to produce a triangle bounded above by the fossa of incus, superficially by the chorda tympani and deeply by the descending portion of the facial nerve, gives access to the round window so that it can be instrumented and a multi-channel cochlear electrode inserted into the scala tympani of the cochlea so that it lies against the spiral lamina and can stimulate the adjacent fibres of the cochlear nerve. This is then connected to the cochlear implant device, which is housed in an extension of the mastoid cavity. Posterior canal obliteration for cupulolithiasis not responsive to Epley's manoeuvre is also undertaken through the mastoid. The endolymphatic sac may be approached by elevating the cortical bone of the anterolateral wall of the posterior cranial fossa, anterior to the sigmoid venous sinus and posterior to the posterior semicircular canal (below a line extended from the axis of the lateral semicircular canal). Decompression of the sac or a shunt procedure is favoured by some for the management of Ménière's disease. Superior canal dehiscences, as seen in Minor's syndrome, are repaired either through the mastoid or by using a middle fossa approach. In patients with intractable Ménière's disease, the whole of the labyrinth can be removed through the mastoid. Vestibular nerve section and the removal of vestibular schwannomas are generally undertaken through retrosigmoid, retrolabyrinthine, translabyrinthine or middle fossa approaches. The choice of approach is dictated or influenced by the size of the tumour, desire or need to preserve hearing, and the surgeon's choice. In recent years, the development of auditory brainstem implants has brought hope to those with profound hearing loss associated with neurofibromatosis type 2 or cochlear nerve aplasia. These devices are similar to a cochlear implant but the electrodes are placed directly into the cochlear nucleus within the brainstem through a retrosigmoid approach.

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